Superpolynomial online mistakes for isotropic Bayesian tanh networks

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Abstract

We construct small Boolean targets and fixed nonrepeating input sequences on which exact Bayesian learning with an ordinary tanh network makes superpolynomially many online mistakes. The network has one hidden layer, a linear readout, and independent Gaussian weights whose covariance is isotropic within each layer. The prior is conditioned from the start on an absolute output margin at every Boolean input. Training conditions on the output sign agreeing with every revealed Boolean label. At every sufficiently large fixed learner width, the posterior probability of the correct label tends to zero on each designated round as the Gaussian scale tends to zero.

The targets have one-hidden-layer tanh teachers of width C, margin one, and coefficients bounded by C. The construction forces $\exp(\Omega(\sqrt{C}/\log C))$ designated mistakes. Its calibration prefix and sufficient learner width can be large. This is an adversarial-order result, rather than an iid sample-complexity lower bound.

The proof first determines the exact optimal affine neuron for a single sufficiently high odd parity. A quantitative support estimate and Fourier localization extend the sharp norm to a transversal code. The optimized scalar costs have positive inverse-degree curvature; their cheapest feasible scores therefore include a fixed negative overshoot. An explicit dual classifies every minimum, a triangular path transfer orders exponentially many competing costs, and integer neuron replication transfers their gaps to the full Gaussian posterior.

1 Model and result

A small circuit may be easy to learn from independent samples while remaining difficult along an adversarial order of inputs. We study that distinction for exact Bayesian prediction with the standard activation tanh. The theorem concerns the posterior distribution, including all tied or nearly tied parameter realizations.

On $X = \{-1, 1\}^d$, append the constant input and write $\bar{x} = (1, x)$. Consider a one-hidden-layer network

$$F(x) = \sum_{j=1}^{N} v_j \tanh(u_j \cdot \bar{x}), \qquad (u_j)_i \sim \mathcal{N}(0, N\sigma^2/d), \qquad v_j \sim \mathcal{N}(0, \sigma^2/N),$$

with independent coordinates. The prior is first conditioned on $|F(x)| > 1/2$ for every input. For a revealed Boolean label y_i, training conditions on $y_i F(x_i) > 1/2$: the output must have the correct sign and margin, while its real value remains free. The strong-prior limit is $\sigma \downarrow 0$ at each fixed width N. Equivalently, at that fixed width set $t = \sqrt{N}\,\sigma$: the variances become t^2/d in the incoming layer

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and t^2/N^2 in the readout. Thus this is the usual strong-prior limit, with isotropy within each layer preserved. Its quadratic energy is

$$K_N = \frac{d}{N}\sum_j \|u_j\|_2^2 + N\sum_j v_j^2.$$

For a function on the finite cube, define the tanh variation cost

$$V(F) = \inf\left\{\sum_j |c_j|\|a_j\|_2 : F(x) = \sum_j c_j \tanh(a_j \cdot \bar{x}) \text{ for all } x\right\}. \tag{1}$$

The infimum is over finite representations and includes the incoming bias coordinate in the norm. The arithmetic–geometric mean inequality always gives

$$K_N \geq 2\sqrt{d}\sum_j |v_j|\|u_j\|_2 \geq 2\sqrt{d}\,V(F). \tag{2}$$

Unlike a homogeneous activation, tanh does not permit arbitrary rescaling of an incoming vector while preserving the neuron output up to a scalar. Equality in (2) at a specified finite width therefore requires a separate argument.

Theorem 1 (Adversarial online mistakes)**.** *For every sufficiently large integer k, put $m = 6k + 6$. There are a dimension $d_m = \Theta(m(\log m)^2)$, a Boolean target f_m on the full cube, and a fixed nonrepeating input sequence with $T_m = 2^{k-1} = \exp(\Omega(m))$ designated rounds such that the following hold. The target has a one-hidden-layer tanh score of margin one and width $C_m = O(m^2(\log m)^2)$, with every coefficient bounded by C_m. It also has a bounded-fan-in Boolean circuit with $O(d_m)$ gates. There is a finite $N_0(m)$ such that, at every fixed $N \geq N_0(m)$, the posterior probability of the correct label tends to zero at each designated round as $\sigma \downarrow 0$. Consequently the sequence forces $\exp(\Omega(\sqrt{C_m}/\log C_m))$ posterior-majority mistakes in the strong-prior limit, and the same number of expected mistakes under posterior sampling.*

Corollary 2 (Finite Gaussian scales)**.** *Fix a construction, any $N \geq N_0(m)$, and $\eta > 0$. For every sufficiently small positive σ, posterior-majority prediction is wrong at all T_m designated rounds, and the correct-label posterior probability is below $1/3$ at every such round. Under fresh posterior sampling, all T_m designated predictions are wrong simultaneously with probability at least $1 - \eta$. The input sequence can be extended to a permutation of the full cube.*

Proof. There are finitely many designated rounds. Choose σ so that the correct-label probability at each is below $\min(1/3, \eta/T_m)$. Majority and the confidence-threshold conclusion follow, while a union bound proves the sampling statement. Append all inputs not yet revealed to extend the sequence without repetition. □

The order of limits is part of the statement: first choose any sufficiently large finite learner width, then send the Gaussian scale to zero. No distribution-shift or typical-order claim is assumed. The input sequence has a potentially large calibration prefix and is independent of the learner width and Gaussian scale. The sufficiently large parity-degree threshold and the width threshold are existential; this proof does not give practical numerical values for them. The main theorem and its finite-scale prediction consequences have been formalized in Lean, using the actual Gaussian posterior and only Lean's standard axioms. The formalization proves every ingredient needed for that result; it does not separately export every stronger standalone appendix assertion. The accompanying verification record gives the paper-to-Lean correspondence and reproduction commands.

All Fourier coefficients use the uniform input law. For a set A, write $\chi_A(x) = \prod_{i\in A} x_i$ and $\widehat{F}(A) = \mathbb{E}[\chi_A F]$. A sharp inequality of the form

$$|\widehat{\tanh(u\cdot\bar{x})}(A)| \le L_A\|u\|_2$$

gives the lower bound $V(\chi_A) \ge L_A^{-1}$. Signed averaging over input flips gives a matching parity representation whenever a neuron attains the sharp correlation ratio. Controlling a sum of several parity coefficients is stronger and is needed for the online construction.

2 Balanced odd-parity atoms

Let $n = 2r+1$ be odd and let $X_1, \dots, X_n$ be independent uniform signs. Put $S = \sum_i X_i$, $\chi = \prod_i X_i$, and

$$q_n(a) = (-1)^r \mathbb{E}[\chi \tanh(aS)], \qquad \kappa_n = 2^{1-n}\binom{n-1}{(n-1)/2}.$$

The constant κ_n is the absolute top Fourier coefficient of majority on n bits. The balanced representation cost is

$$w_n = \inf_{a>0} \frac{a\sqrt{n}}{q_n(a)}. \tag{3}$$

The positivity needed in this definition follows from the next formula.

Lemma 3 (A positive diagonal series)**.** *For every odd $n \ge 1$ and $a > 0$, with $c_\ell = (\ell + 1/2)\pi$,*

$$q_n(a) = 2n!a^n \sum_{\ell\ge 0} \prod_{j=0}^{(n-1)/2} \left(c_\ell^2 + (2j+1)^2a^2\right)^{-1} > 0. \tag{4}$$

Proof. The sine transform of tanh is

$$\tanh t = \int_0^\infty \frac{\sin(t\omega)}{\sinh(\pi\omega/2)}\, d\omega.$$

Averaging the parity character gives

$$q_n(a) = \int_0^\infty \frac{\sin^n(a\omega)}{\sinh(\pi\omega/2)}\, d\omega.$$

Expand $1/\sinh(\pi\omega/2) = 2\sum_{\ell\ge 0} e^{-c_\ell\omega}$. Absolute integrability of $|\sin(a\omega)|^n/\sinh(\pi\omega/2)$ justifies the exchange. Repeated integration by parts gives

$$\int_0^\infty e^{-c\omega}\sin^n(a\omega)\, d\omega = \frac{n!a^n}{\prod_{j=0}^{(n-1)/2}(c^2 + (2j+1)^2a^2)}.$$

This proves the formula and positivity. □

Theorem 4 (Unique balanced gain and full local optimality)**.** *For every odd $n \ge 5$, the ratio $q_n(a)/a$ has a unique global maximum at a gain $a_n \in (1,2)$. Moreover*

$$J_n(b,u) = \frac{(-1)^{(n-1)/2}\mathbb{E}[\chi\tanh(b + u\cdot X)]}{\sqrt{b^2 + \|u\|_2^2}}$$

has a strict local maximum at $(b,u) = (0, a_n\mathbf{1})$. Its Hessian there is negative definite. The absolute Fourier-correlation ratio equals J_n in a neighborhood of that point.

Proof. For any odd function h, pairing opposite Hamming levels yields

$$(-1)^r \mathbb{E}[\chi h(aS)] = 2^{1-n} \sum_{j=0}^{r} (-1)^j \binom{n}{r-j} h(a(2j+1)). \tag{5}$$

Let $p(t) = -\tanh'' t = 2 \tanh t \operatorname{sech}^2 t$. It is positive and strictly decreasing for $t \geq 1$. Consequently

$$D_n(a) := (-1)^r \mathbb{E}[\chi \tanh''(aS)] < 0 \qquad (a \geq 1) \tag{6}$$

for every odd $n \geq 1$: the positive terms in the corresponding alternating sum strictly decrease.

The same argument proves $q_n''(a) < 0$ on $a \geq 1$. The extra factor in its terms is $(2j+1)^2$. For $t' = t + 2a$, $t \geq a \geq 1$,

$$\frac{p(t')}{p(t)} \leq e^{-4a} \frac{(1+e^{-2a})^3}{1-e^{-2a}}.$$

The ratio of the squared integer factors is at most nine. Since $e^2 > 7$,

$$9e^{-4a} \frac{(1+e^{-2a})^3}{1-e^{-2a}} < \frac{9}{49} \frac{(8/7)^3}{6/7} < 1.$$

The binomial factors decrease as well, giving the claimed strict sign.

For $0 < a \leq 1$, every summand of $q_n(a)/a$ in (4) has positive derivative. Its logarithmic derivative multiplied by a is

$$-2 + 2 \sum_{j=0}^{r} \frac{c_\ell^2}{c_\ell^2 + (2j+1)^2 a^2}.$$

Using $n \geq 5$ and $c_\ell^2 \geq \pi^2/4 > 12/5$, the sum is greater than

$$\frac{12}{17} + \frac{12}{57} + \frac{12}{137} = \frac{44428}{44251} > 1.$$

The differentiated series is locally uniformly convergent on $a > 0$, so $(q_n(a)/a)' > 0$ throughout $(0, 1]$.

Put $H_n(a) = a q_n'(a) - q_n(a)$. Its derivative is $a q_n''(a) < 0$ for $a \geq 1$. Define

$$g(t) = t \operatorname{sech}^2 t + 1 - \tanh t, \qquad g'(t) = -2t \operatorname{sech}^2 t \tanh t < 0.$$

Subtracting the majority coefficient in (5), and using alternating-sum bounds, gives

$$H_n(1) \geq -\kappa_n + B_0[g(1) - g(3)], \qquad H_n(2) \leq -\kappa_n + B_0 g(2),$$

where $B_0 = 2^{1-n}\binom{n}{r}$ and $B_0/\kappa_n = 2n/(n+1)$. The elementary bounds $7 < e^2 < 37/5$ imply

$$g(1) > \frac{290}{441}, \quad g(3) < \frac{14}{343}, \quad g(1) - g(3) > \frac{3}{5}, \quad g(2) < \frac{10}{49} < \frac{1}{2}.$$

As $5/3 \leq B_0/\kappa_n < 2$, we obtain $H_n(1) > 0$ and $H_n(2) < 0$. Strict decrease proves the unique global maximizing gain.

To check the affine Hessian, write $R = a_n\sqrt{n}$ and $q = q_n(a_n) > 0$. Permutation symmetry splits weight directions into $\operatorname{span}(\mathbf{1})$ and its orthogonal complement. Bias–weight mixed derivatives vanish under $X \mapsto -X$. The bias eigenvalue of J_n is

$$\frac{D_n(a_n)}{R} - \frac{q}{R^3} < 0.$$

For a unit difference direction $(e_i - e_j)/\sqrt{2}$, conditioning on $X_i = -X_j$ shows that the numerator's second derivative is $D_{n-2}(a_n)$. Thus the common eigenvalue on $\mathbf{1}^\perp$ is

$$\frac{D_{n-2}(a_n)}{R} - \frac{q}{R^3} < 0.$$

Along the balanced line, at the stationary gain,

$$\frac{d^2}{da^2}\frac{q_n(a)}{a\sqrt{n}}\bigg|_{a=a_n} = \frac{q_n''(a_n)}{a_n\sqrt{n}} < 0.$$

Rescaling this direction to unit length preserves its sign. The gradient vanishes by the stationary condition and symmetries. All Hessian blocks are therefore negative definite. □

Corollary 5 (A local weighted-code inequality)**.** *Partition physical coordinates into m disjoint pairs of rails (B_i^0, B_i^1), where m is odd and each rail has odd size at least three. For $s \in \mathbb{F}_2^m$ put $A_s = \bigcup_i B_i^{s_i}$ and $n_s = |A_s|$, and assume $n_s \geq 5$. Define*

$$Q(b,u) = \frac{\sum_s w_{n_s} |\mathbb{E}[\chi_{A_s} \tanh(b + u \cdot X)]|}{\sqrt{b^2 + \|u\|_2^2}}.$$

At every point with $b = 0$, incoming weights a_{n_s} on one selected transversal A_s, and zero weights elsewhere, Q has value one and a strict local maximum. Input flips give the corresponding signed points.

Proof. The selected parity coefficient has the negative-definite quadratic expansion in Theorem 4. Every inactive coordinate has the same numerator second derivative as the bias, and mixed inactive coordinates vanish by independence. Hence their normalized eigenvalue is also $D_n/R - q/R^3 < 0$.

Each other transversal contains at least three inactive coordinates. Repeated finite differences factor its coefficient by the product of those incoming weights, so its absolute coefficient is $O(\|(\Delta b, \Delta u)\|^3)$ near the selected point. The finite sum of these terms does not change the quadratic part. The selected weight w_n makes its value exactly one. We use this second-order expansion; the absolute-value objective need not be twice differentiable throughout a neighborhood. □

Corollary 5 does not assert $Q \leq 1$ globally. A violating neuron could be far from every selected balanced atom.

3 The sharp global bound for three-bit parity

For three bits the global problem admits an exact algebraic solution. Write

$$C_3(u) = \mathbb{E}[X_1 X_2 X_3 \tanh(u \cdot X)], \qquad q_3(a) = \frac{2\tanh^3 a}{1 + 3\tanh^2 a}.$$

Theorem 6 (Global three-bit parity gain)**.** *There is a unique maximizer $a_3 > 0$ of $q_3(a)/a$. Put*

$$L_3 = \frac{q_3(a_3)}{\sqrt{3}\, a_3}.$$

Every affine tanh neuron, including neurons with additional irrelevant input coordinates, satisfies

$$|\widehat{\tanh(u \cdot \bar{X})}(\{1,2,3\})| \leq L_3 \|u\|_2.$$

Nonzero equality requires zero bias, zero irrelevant weights, and equal absolute relevant weights a_3. Consequently the variation cost of three-bit parity is exactly L_3^{-1}.

Proof. First assume positive relevant weights a, b, c and no bias. Let $t_1 = \tanh a, t_2 = \tanh b, t_3 = \tanh c$ and $s_i = t_i^2$. Direct use of the addition formula gives

$$|C_3(a,b,c)| = \frac{2t_1t_2t_3\prod_i(1-s_i)}{P},$$

where

$$P = 1 - 2\sum_{i<j} s_is_j + \sum_{i<j} s_i^2s_j^2 - 2s_1s_2s_3\sum_i s_i + 8s_1s_2s_3.$$

The denominator obeys the identity

$$P - \prod_i(1-s_i)\left(1+\sum_i s_i\right) = \frac{1}{2}\sum_{\text{cyc}}(s_i-s_j)^2(1-s_k)^2 \geq 0.$$

Thus, by the arithmetic–geometric mean inequality,

$$|C_3(a,b,c)| \leq \frac{2t_1t_2t_3}{1+s_1+s_2+s_3} \leq \prod_{i=1}^{3}\left(\frac{2t_i^3}{1+3t_i^2}\right)^{1/3} = \prod_i q_3(u_i)^{1/3}. \tag{7}$$

Equality with positive weights requires all three equal. Input flips handle negative weights, and a zero relevant weight makes the coefficient zero. Applying $q_3(a) \leq a\sup_t q_3(t)/t$ and then geometric mean at most quadratic mean proves the sharp bias-free bound.

For uniqueness of the scalar gain, set $t = \tanh a$. The sign of $(q_3(a)/a)'$ is the sign of

$$H(t) = 3\operatorname{arctanh}(t)(1-t^4) - t(1+3t^2).$$

Here $H(0) = 0$, $H'(0) = 2$, $H(1^-) = -4$, and

$$H'(t) = 2 - 6t^2 - 12t^3\operatorname{arctanh} t, \qquad H''(t) < 0 \quad (0 < t < 1).$$

Strict concavity gives exactly one zero in $(0,1)$, proving uniqueness.

We record the bias argument generally. For any odd activation and odd n, its top parity coefficient satisfies

$$C_n(B;u) = \frac{1}{2}\left[C_n(0;u_1,\ldots,u_{n-1},u_n+B) + C_n(0;u_1,\ldots,u_{n-1},u_n-B)\right]. \tag{8}$$

Indeed, averaging the first $n-1$ coordinates with their parity gives an odd function of the remaining scalar offset. If the bias-free gain is at most $L\|u\|$, triangle inequality and root mean square yield

$$|C_n(B;u)| \leq \frac{L}{2}\left[\sqrt{A+(u_n+B)^2} + \sqrt{A+(u_n-B)^2}\right] \leq L\sqrt{A+u_n^2+B^2},$$

where $A = \sum_{i<n} u_i^2$. For a nonzero coefficient, $u_n \neq 0$, so equality in root mean square forces $B = 0$. Conditioning on irrelevant coordinates and applying the same argument puts their squared norm into the bias second moment. A nonzero irrelevant weight makes the effective bias nonzero with positive probability and prevents equality.

Finally, signed averaging over the eight input flips projects a balanced maximizing neuron to $q_3(a_3)\chi_{\{1,2,3\}}$, up to its fixed sign. This gives a finite representation of parity with cost $\sqrt{3}a_3/q_3(a_3)$, matching the lower bound from the sharp gain. □

Numerically $a_3 \approx 0.9655523$ and $L_3^{-1} \approx 5.3674918$. The numerical values are not used in the proof. The eventual high-degree global theorem below uses a different argument; it does not require a higher-degree version of (7).

4 Positive curvature of the balanced parity cost

Theorem 7 (Optimized cost expansion)**.** *Along odd integers $n \to \infty$, the balanced costs satisfy*

$$w_n = An + B + \frac{C}{n} + O(n^{-2}), \qquad C > 0. \tag{9}$$

They admit a holomorphic interpolation on $\{\Re z > 0,\ |z| > z_0\}$ with this uniform expansion. In particular,

$$w''(n) = \frac{2C}{n^3} + O(n^{-4}).$$

Numerically $A \approx 2.24178537487$, $B \approx -1.52970827506$, and $C \approx 0.59356404999$.

Proof. Write $J_j = j(j+1)$ and

$$R_j(z) = \prod_{\ell=0}^{j-1} \frac{z-(2\ell+1)}{z+(2\ell+3)}, \qquad f_j(a) = \frac{1}{1+e^{2a(2j+1)}}.$$

Pairing Hamming levels in (5) gives the exact identity

$$q_n(a) = \kappa_n F(n,a), \qquad F(z,a) = 1 - \frac{4z}{z+1}\sum_{j\geq 0}(-1)^j R_j(z) f_j(a). \tag{10}$$

At odd n, $R_j(n) = 0$ for $j > (n-1)/2$. In the right half-plane each factor of R_j has modulus at most one, since

$$|z+(2\ell+3)|^2 - |z-(2\ell+1)|^2 = 8(\ell+1)(\Re z+1) > 0.$$

The series is therefore locally normally convergent for $\Re a > 0$. Its large-z expansion is

$$F(z,a) = F_0(a) + F_1(a)/z + F_2(a)/z^2 + O(|z|^{-3}),$$

where

$$\begin{aligned} F_0 &= 1 - 4\sum_{j\geq 0}(-1)^j f_j, \\ F_1 &= 4\sum_{j\geq 0}(-1)^j (1+2J_j) f_j, \\ F_2 &= -4\sum_{j\geq 0}(-1)^j (1+4J_j+2J_j^2) f_j. \end{aligned}$$

For completeness, expand

$$R_j(z) = 1 - 2J_j/z + (2J_j + 2J_j^2)/z^2 + O_j(|z|^{-3}).$$

To sum the remainder uniformly, split at $j = \eta|z|$ for a sufficiently small fixed η. Below the cutoff, the logarithmic Taylor remainder is a polynomial in j times $|z|^{-3}e^{O(j^2/|z|)}$; its exponential is absorbed by f_j's $e^{-4\Re(a)j}$ decay. Above the cutoff, $|R_j| \leq 1$ makes the tail exponentially small. This argument holds uniformly near the real interval $[1.237, 1.238]$, including any fixed number of a-derivatives.

The gamma-ratio interpolation of κ_n has uniform expansion

$$\kappa_z = \sqrt{\frac{2}{\pi z}}\left(1 + \frac{1}{4z} + \frac{1}{32z^2} + O(|z|^{-3})\right) \qquad (|\arg z| \leq \pi/2).$$

Set $D_1 = F_1 + F_0/4$ and $D_2 = F_2 + F_1/4 + F_0/32$. Then

$$\frac{a\sqrt{z}}{\kappa_z F(z,a)} = A(a)z + B(a) + C_{\mathrm{raw}}(a)/z + O(|z|^{-2}),$$

with

$$A(a) = \sqrt{\pi/2}\,\frac{a}{F_0}, \qquad B(a) = -A(a)D_1/F_0,$$
$$C_{\mathrm{raw}}(a) = A(a)\left[(D_1/F_0)^2 - D_2/F_0\right].$$

Let a_0 be the zero in $[1.237, 1.238]$ of $aF_0' - F_0$. The exact certificate below proves $F_0'' < 0$ there, so $A''(a_0) > 0$. The analytic implicit-function theorem gives a critical branch $a(z)$ near a_0 for all sufficiently large z in the right half-plane. For real odd n, Theorem 4's uniqueness identifies it with a_n. Substitution gives

$$A = A(a_0), \quad B = B(a_0), \quad C = C_{\mathrm{raw}}(a_0) - \frac{B'(a_0)^2}{2A''(a_0)}. \tag{11}$$

This proves the claimed analytic remainder. Cauchy's estimate on a radius-$n/2$ disk gives the second-derivative expansion.

Here are reproducible exact bounds for the sign in (11). At a_0, using $F_0' = F_0/a_0$,

$$\frac{C}{\sqrt{\pi/2}} = \frac{aD_1^2}{F_0^3} - \frac{aD_2}{F_0^2} + \frac{(D_1 - aD_1')^2}{2aF_0''F_0^2}. \tag{12}$$

The companion script `verification/certify_curvature.py` uses only rational arithmetic in its assertions. It encloses exponentials by their Taylor sums through degree 256 and a geometric bound on the remaining positive terms. It sums the six series $F_0, F_1, F_2, F_0', F_0'', F_1'$ through $j = 9$. On $[1237/1000, 619/500]$, each omitted tail is bounded by

$$192 \cdot 11^6 e^{-2(1237/1000)21} < 10^{-14}.$$

Indeed, every polynomial factor is bounded by $16(j+1)^6$, with the common factor four, and a geometric ratio at most $64e^{-4(1237/1000)} < 64/100$; the sum of this geometric series is less than three. Outward interval rounding uses denominator 10^{45}. The resulting rational enclosures prove opposite signs of $aF_0' - F_0$ at the endpoints, $F_0'' < 0$ throughout the interval, and

$$0.4616 < C/\sqrt{\pi/2} < 0.4842.$$

Thus $C > 0$. Floating-point formatting is used only to display the certificate's already proved interval endpoints. □

5 Exact tanh code norm from support stability and high-order absorption

Write $h_\theta(x) = \tanh(b + u \cdot x)$ and $r = \|\theta\|_2 = (b^2 + \|u\|_2^2)^{1/2}$, with uniform Boolean input. For an odd set A of cardinality n, let $c_A(\theta) = \widehat{h_\theta}(A)$. The global single-parity theorem (Theorem 31) supplies the positive number w_n such that

$$w_n\,|\mathbb{E}\chi_A(X)\tanh(u_A \cdot X)| \le \|u_A\|_2. \tag{13}$$

This holds for every sufficiently large odd n, with equality at the balanced optimal-gain atom. Its scalar estimates also give $w_n \le C_w n$ for an absolute constant C_w. The argument below requires no quantitative stability of the balanced gain or of the selected coordinates.

Lemma 8 (Global support stability)**.** *Assume* (13) *for an odd support* A*. Set*

$$v_A = b^2 + \|u_{A^c}\|_2^2, \qquad \Delta_A = r - w_n|c_A(\theta)|, \qquad S_n = \max\{2, 96\sqrt{6}\, w_n^2\}.$$

Then $\Delta_A \geq 0$ *and*

$$v_A \leq S_n r \Delta_A. \tag{14}$$

Consequently the squared L^2 *distance from* h_θ *to the space of odd functions depending only on* x_A *is at most* $S_n r \Delta_A$*.*

Proof. Fix any selected coordinate of weight a, put $U = \|u_A\|_2^2$, and condition on the outside coordinates, writing $B = b + u_{A^c} \cdot X_{A^c}$. Oddness of n and of tanh gives the exact identity

$$c_A(B; u_A) = \tfrac{1}{2}\left[c_A(0; u_A + Be_i) + c_A(0; u_A - Be_i)\right].$$

Applying (13) and averaging yields

$$w_n|c_A(\theta)| \leq \mathbb{E}\frac{\sqrt{U + B^2 + 2aB} + \sqrt{U + B^2 - 2aB}}{2} \leq \mathbb{E}\sqrt{U + B^2} \leq r.$$

For $x \geq |t|$, concavity with its quadratic remainder gives

$$\sqrt{x} - \frac{\sqrt{x+t} + \sqrt{x-t}}{2} \geq \frac{t^2}{8x^{3/2}}.$$

For example, divide by $\sqrt{x}$ and use $1 - \frac{1}{2}(\sqrt{1+z} + \sqrt{1-z}) \geq z^2/8$ on $[-1, 1]$. Thus

$$\Delta_A \geq \frac{a^2}{2}\,\mathbb{E}\frac{B^2}{(U + B^2)^{3/2}}. \tag{15}$$

Here $\mathbb{E}B^2 = v_A$ and $\mathbb{E}B^4 \leq 3v_A^2$. If $v_A > 0$, then

$$\mathbb{E}[B^2 \mathbf{1}_{B^2 \leq 6v_A}] \geq v_A/2, \qquad U + 6v_A \leq 6r^2,$$

so (15) implies $\Delta_A \geq a^2 v_A/(24\sqrt{6}\, r^3)$. The cases $r = 0$ or $v_A = 0$ are immediate. If $w_n|c_A| < r/2$, the bound $v_A \leq r^2 \leq 2r\Delta_A$ suffices. Otherwise the one-coordinate finite-difference formula and the one-Lipschitz property of tanh give $|a| \geq |c_A| \geq r/(2w_n)$, and hence $v_A \leq 96\sqrt{6}\, w_n^2 r \Delta_A$. Finally $h_A(x) = \tanh(u_A \cdot x_A)$ is odd and depends only on A, while

$$\|h_\theta - h_A\|_{L^2}^2 \leq \mathbb{E}(b + u_{A^c} \cdot X_{A^c})^2 = v_A.$$

□

There are $p \geq 2$ pairs of disjoint even rails, each of size in $[R, R + \Delta]$, where $R \geq 4$ is even and $\Delta \leq R$. There is also one shared carrier coordinate. A transversal support consists of the carrier and one entire rail from each pair. Let $\mathcal{C}$ be any nonempty retained family of such supports, and put

$$D = pR, \qquad W = \max_{A \in \mathcal{C}} w_{|A|}, \qquad Q(\theta) = \sum_{A \in \mathcal{C}} w_{|A|}|c_A(\theta)|, \qquad S = \max\{2, 96\sqrt{6}\, W^2\}.$$

Every support has odd size between $D + 1$ and $2D + 1$, so $W = O(D)$ and $S = O(D^2)$. Let $\mathcal{D}_{\text{odd}}$ be the linear span of Walsh characters indexed by odd subsets of retained supports.

Theorem 32 supplies exactly the following quantitative orientation bound. There is an absolute C_0 such that, for every prescribed $L > 0$, there is K_L for which all sufficiently large p and all $R \geq K_L \log^2(p+2)$ satisfy: whenever $Q(\theta) \geq r/2$ and $r > 0$, there is $A \in \mathcal{C}$ for which

$$r \leq C_0\sqrt{D}, \qquad E_A := \sum_{B \in \mathcal{C}\setminus\{A\}} w_{|B|}|c_B(\theta)| \leq D^{-L}r. \tag{16}$$

Fix $K_0 = K_8$ for the remainder of this section. The sufficiently-large-p threshold is uniform in R, the retained family, the ambient dimension, and the atom. A lower bound on $r/\sqrt{D}$ also follows from that lemma but is not needed here.

Lemma 9 (Local vanishing of other orientations)**.** *For any retained A, put $\tau = \|u_{A^c}\|_2$ and $\rho = D^{-2}$. If $\tau \leq \rho$, then, for all sufficiently large p,*

$$E_A \leq A_{p,R}\tau^2, \qquad A_{p,R} = 2WpD^4(4D^{-3/2}/\pi)^R. \tag{17}$$

Uniformly for $R \geq K_0 \log^2(p+2)$, as $p \to \infty$,

$$A_{p,R} \leq \frac{1}{2SC_0\sqrt{D}}. \tag{18}$$

Proof. For real t the disk $|z - t| \leq \pi/4$ lies in a strip on which $|\tanh z| \leq 1$. Cauchy's estimate gives $\sup_{t\in\mathbb{R}} |\tanh^{(k)} t| \leq k!(4/\pi)^k$. A transversal B switching j pairs relative to A contains at least jR coordinates outside A. Choose exactly jR of them. Repeated centered differences and AM–GM give, since $jR \leq D$,

$$|c_B(\theta)| \leq (jR)!(4/\pi)^{jR} \prod_{i \text{ chosen}} |u_i| \leq (4\sqrt{jR}\,\tau/\pi)^{jR} \leq (4\sqrt{D}\,\tau/\pi)^{jR}.$$

There are at most $\binom{p}{j}$ such transversals. Because $jR \geq 2$, $\tau^{jR} \leq \rho^{jR-2}\tau^2$, and therefore

$$E_A \leq W\tau^2\rho^{-2}\left[(1 + (4\sqrt{D}\rho/\pi)^R)^p - 1\right].$$

The quantity $p(4D^{-3/2}/\pi)^R$ tends uniformly to zero, so $(1+x)^p - 1 \leq 2px$ gives (17). Its logarithm is bounded by

$$O(\log D + \log p) - \tfrac{3}{2}R\log D + O(R).$$

This is smaller than any fixed negative multiple of $\log D$, uniformly for $R \geq K_0 \log^2(p+2)$, as $p \to \infty$. Since $S = O(D^2)$, this proves (18). □

Theorem 10 (Exact weighted code norm and polynomial support stability)**.** *For all sufficiently large p, uniformly for even $R \geq K_0 \log^2(p+2)$, $\Delta \leq R$, every retained family, every ambient cube, and every affine tanh atom,*

$$Q(\theta) \leq r, \tag{19}$$

and

$$\|\Pi_{\mathcal{D}_{\text{odd}}^{\perp}} h_\theta\|_{L^2}^2 \leq 2Sr(r - Q(\theta)). \tag{20}$$

The norm bound is sharp: every balanced optimal-gain atom on one retained support attains equality. A nonzero equality atom has zero bias and zero weights outside a single retained support; the intrinsic equality theorem then identifies its selected weights as a balanced optimal-gain atom.

Proof. The zero atom is immediate. If $Q < r/2$, the norm assertion is immediate, and $|\tanh t| \le |t|$ gives $\|h_\theta\|_2^2 \le r^2 \le 2r(r-Q)$, proving the stated stability bound. Otherwise choose A from (16) with $L = 8$ and put $\Delta_A = r - w_{|A|}|c_A| \ge 0$. The exact identity is

$$r - Q = \Delta_A - E_A.$$

We claim $E_A \le \Delta_A/2$. This is immediate if $\Delta_A \ge 2E_A$. In the remaining case, support stability and (16) imply

$$v_A \le Sr\Delta_A < 2SrE_A \le 2SC_0^2 D^{-7} = O(D^{-5}).$$

For sufficiently large p this is at most $D^{-4} = \rho^2$, uniformly in R. Thus $\tau^2 \le v_A \le \rho^2$, and (18) together with $r \le C_0\sqrt{D}$ yields

$$E_A \le \frac{\tau^2}{2SC_0\sqrt{D}} \le \frac{v_A}{2Sr} \le \frac{\Delta_A}{2}.$$

This establishes the claim in all cases, and hence $r - Q \ge \Delta_A/2 \ge 0$. Moreover the selected bias-free atom lies in $\mathcal{D}_{\mathrm{odd}}$, so

$$\|\Pi_{\mathcal{D}_{\mathrm{odd}}^\perp} h_\theta\|_2^2 \le v_A \le Sr\Delta_A \le 2Sr(r-Q).$$

If equality holds in the norm bound, then $\Delta_A = 0$ and $v_A = 0$. Conversely a balanced optimal-gain atom on A has its sharp single-parity coefficient, and every other transversal coefficient vanishes because that transversal contains coordinates outside A. This proves sharpness and the equality assertions. □

5.1 Resource consequence

The constant in (20) is $2S = O(D^2)$, with an absolute implied constant. The theorem now needs only a polylogarithmic baseline rail size. Larger effective logical increments can be split among repeated physical columns, each of increment at most R, preserving the hypotheses. The scalar construction below uses $p = \Theta(m)$, $R = \Theta(\log^2 m)$ and $D = \Theta(m \log^2 m)$. The odd downward space is essential: the full downward space would also contain the even character obtained by deleting the carrier from an entire transversal, which is not covered by calibration observability.

6 An exact repair of the logical score optimization

The output magnitudes are free optimization variables even when all labels are fixed. For the scalar weights below, the cheapest competitors use a negative overshoot on one fixed logical coset. We first identify that optimizer in a finite-dimensional weighted Fourier problem, then establish the scalar estimates used by the neural construction.

For a score g on $G = \mathbb{F}_2^m$, write $B_w(g) = \sum_s w_s|\widehat{g}(s)|$ with normalized Walsh coefficients. The sharp atomic theorem identifies this weighted norm with the variation cost on the retained physical code by signed input-flip averaging of its equality atoms.

6.1 A reduction to nonnegative Fourier coefficients

Let $G = \mathbb{F}_2^m$, $M = |G|$, and let $w_s \ge w_0 > 0$. Continue to write $B_w(g) = \sum_s w_s|\widehat{g}(s)|$ and $W(x) = M^{-1}\sum_s w_s(-1)^{s\cdot x}$.

Lemma 11 (The point-positive margin problem)**.** *Suppose $M \ge 4$. Among scores satisfying $g(0) \ge 1/2$ and $g(x) \le -1/2$ for every $x \ne 0$, there is a minimum-cost score with $g(0) = 1/2$, $\widehat{g}(0) < 0$, and $\widehat{g}(s) \ge 0$ for $s \ne 0$. If $w_s > w_0$ for every $s \ne 0$, every optimizer has these properties.*

Proof. First impose $c_s = \widehat{g}(s) \geq 0$ for $s \neq 0$ and $c_0 = 1/2 - \sum_{s \neq 0} c_s$. The margin constraints become

$$\sum_{s : s \cdot x = 1} c_s \geq \frac{1}{2} \quad (x \neq 0),$$

and their sum implies $\sum_{s \neq 0} c_s \geq 1 - 1/M$, hence $c_0 < 0$. The resulting objective is $-w_0/2 + \sum_{s \neq 0} (w_0 + w_s) c_s$. This is a feasible, attained finite linear program. Its dual has variables $a_x \geq 0$ and constraints

$$\sum_{x : s \cdot x = 1} a_x \leq w_0 + w_s,$$

with objective $\frac{1}{2} \sum_{x \neq 0} a_x - w_0/2$. For an optimal dual, define a function λ by

$$\lambda(0) = M \left(\frac{1}{2} \sum_{x \neq 0} a_x - w_0 \right), \qquad \lambda(x) = -M a_x / 2 \quad (x \neq 0).$$

Then $\widehat{\lambda}(0) = -w_0$ and

$$\widehat{\lambda}(s) = -w_0 + \sum_{x : s \cdot x = 1} a_x \in [-w_0, w_s].$$

Consequently $|\widehat{\lambda}(s)| \leq w_s$. The restricted optimum is at least $(3/2 - 2/M) w_0 > w_0/2$, so $\lambda(0) > 0$. Thus λ is also a dual certificate for the unrestricted margin problem and has the same value. Equality forces $g(0) = 1/2$ and $c_0 \leq 0$. Under strict $w_s > w_0$, it also excludes negative c_s for $s \neq 0$; the summed constraints then give $c_0 < 0$. □

If the prescribed positive set is a subgroup $J \leq G$, averaging over translations by J deletes all Fourier coefficients outside $J^\perp$. It strictly decreases the norm whenever any such coefficient is nonzero. Thus every optimizer is J-invariant, and the problem reduces to the point-positive problem on G/J with weights restricted to $J^\perp$.

6.2 A dominant Walsh coefficient selects a unique overshoot

Theorem 12 (Exact subgroup-positive optimizer)**.** *Let $J \leq G$, let $u \notin J$, and set $H = \langle J, u \rangle$, $j = |J|$, and $L = M/(2j) \geq 2$. Assume $w_s > w_0 > 0$ for $s \neq 0$. On G/J put*

$$A_x = \sum_{a \in J} W(x + a), \qquad t = \frac{2 j w_0}{M}, \qquad h_x = A_x - t, \qquad h = h_u > 0.$$

For every nonzero coset $c \in G/H$, choose $x \in c$ and put

$$\kappa_c = \max(0, A_x, A_{x+u}), \qquad r_c = \frac{\kappa_c}{h}, \qquad R_* = \sum_{c \neq 0} r_c.$$

These definitions do not depend on the representatives. If

$$R_* < 1, \qquad w_0 > (1 + R_*)(h + t), \tag{21}$$

then the unique minimum of $B_w(g)$ over the full-margin sign cell positive exactly on J is

$$g_J(x) = -\tfrac{1}{2} + \mathbf{1}_{x \in J} - \mathbf{1}_{x \in u + J}. \tag{22}$$

Its cost is

$$C_J = \frac{w_0}{2} + A_0 - A_u. \tag{23}$$

Proof. Work on G/J. By Lemma 11, every optimizer has value $1/2$ at the origin and nonnegative nonconstant Fourier coefficients. It can therefore be written uniquely as

$$g = -\tfrac{1}{2} + \mathbf{1}_0 - \sum_{x \neq 0} \nu_x \mathbf{1}_x, \qquad \nu_x \geq 0.$$

Nonnegativity of its nonconstant coefficients is exactly

$$\sum_{x \neq 0} \nu_x (-1)^{s \cdot x} \leq 1 \quad (s \in J^\perp,\ s \neq 0). \tag{24}$$

Substitution in the objective gives the equivalent maximization

$$C_J = \frac{w_0}{2} + A_0 - t - \max_{\nu} \sum_{x \neq 0} \nu_x h_x.$$

We certify that the unique maximizer is $\nu_u = 1$.

Define a probability law on $H^\perp$ by

$$P(s) = \frac{1}{L}\left[1 + \sum_{c \neq 0} r_c (-1)^{s \cdot x_c}\right].$$

It is strictly positive because $R_* < 1$, and character orthogonality shows that it sums to one. Write $p = P(0) = (1 + R_*)/L$ and condition on $s \neq 0$ to obtain Q. The second condition in (21) implies $p < 1$. For $x \notin H$,

$$\mathbb{E}_Q (-1)^{s \cdot x} = \frac{r_{[x]} - p}{1 - p} \geq \frac{A_x}{h} - \frac{p}{1 - p} > \frac{A_x - t}{h}.$$

The strict last inequality is equivalent to $w_0 > (1 + R_*)(h + t)$, using $t = w_0/L$. For $x = u$ modulo J, the character is identically one and $h_x/h = 1$. Averaging (24) against Q now gives

$$\sum_{x \neq 0} \nu_x h_x \leq h \mathbb{E}_Q \sum_{x \neq 0} \nu_x (-1)^{s \cdot x} \leq h.$$

Equality requires $\nu_x = 0$ outside $u + J$; the remaining constraint forces $\nu_u \leq 1$. The feasible choice $\nu_u = 1$ attains equality, proving uniqueness and both displayed formulas. □

6.3 Walsh signs from the inverse Laplace kernel

The scalar construction only needs a positive inverse Laplace kernel near zero. No derivative estimate whose order grows with the logical dimension is required. Lemma 15, applied to Theorem 7, gives

$$w(n) = An + B + \int_0^\infty e^{-nt} \rho(t)\, dt,$$

where, for fixed positive constants, $c_0 \leq \rho(t) \leq C_0$ on $(0, t_0]$ and $|\rho(t)| \leq C_1 e^{c_1 t}$ everywhere.

Lemma 13 (Uniform high-order Walsh signs)**.** *Consider an auxiliary increment cube with q coordinates, increments $\delta_i \geq 2$, total increment $D_\delta = \sum_i \delta_i$, and $n_s = n_0 + \sum_i \delta_i s_i$. Along a sequence with $n_0 \to \infty$, assume*

$$D_\delta = o(n_0), \qquad q \log(n_0 + D_\delta) = o(n_0).$$

For every support T of size $2 \leq k \leq q$, uniformly in T,

$$\frac{c \prod_{i \in T} \delta_i}{2^k (n_0 + D_\delta)^{k+1}} - E_0 \leq W(T) \leq \frac{C k! \prod_{i \in T} \delta_i}{2^k n_0^{k+1}} + E_0, \qquad E_0 = O(e^{-c' n_0}). \tag{25}$$

In particular all these coefficients are strictly positive for all sufficiently large construction parameters. Singleton coefficients are negative.

Proof. The affine part $An + B$ has zero coefficient at $|T| \geq 2$. Factoring the Laplace integrand gives

$$W(T) = \int_0^\infty e^{-n_0 t} \rho(t) \prod_{i \in T} \frac{1 - e^{-\delta_i t}}{2} \prod_{i \notin T} \frac{1 + e^{-\delta_i t}}{2} \, dt.$$

All factors except ρ lie in $[0, 1]$. For $a, t > 0$,

$$(at/2)e^{-at} \leq (1 - e^{-at})/2 \leq at/2, \qquad e^{-at} \leq (1 + e^{-at})/2 \leq 1.$$

Put $N = n_0 + D_\delta$. On $[1/N, 2/N] \subset (0, t_0]$, these inequalities give a positive contribution at least $c_0 e^{-2} \prod_{i \in T} \delta_i / (2^k N^{k+1})$. The potentially adverse tail on $t \geq t_0$ is bounded in absolute value by $E_0 = O(e^{-c' n_0})$. This proves the lower bound. For the upper bound, use $\rho \leq C_0$ near zero and extend the nonnegative majorant $t^k e^{-n_0 t}$ to the full half-line. Its gamma integral is $k!/n_0^{k+1}$, giving the upper bound. Since $\delta_i \geq 2$, the positive lower contribution is at least $cN^{-(q+1)}$, whose logarithm is $-o(n_0)$ by hypothesis. It therefore dominates E_0 uniformly, proving positivity. Finally $w'(a) = A + O(a^{-2}) > 0$ for large a, and the first-difference formula proves the singleton assertion. □

Lemma 14 (A dominant pair of increments)**.** *Under Lemma 13, let two distinguished increments belong to $[H, 2H]$ and put $S = \sum_{j \geq 3} \delta_j$. If $S = o(H)$ and $H = o(n_0)$, then, for $u = e_1 + e_2$,*

$$\frac{\sum_{|T| \geq 2,\ T \neq u} W(T)}{W(u)} = O(S/H + H/n_0). \tag{26}$$

Proof. The lower bound gives $W(u) \geq cH^2/n_0^3$ for large parameters. Other degree-two increment products sum to $O(HS + S^2)$, so their upper bounds divided by $W(u)$ give $O(S/H)$, apart from tails. Write $D_\delta = O(H)$. For $k \geq 3$ use $k! \sum_{|T|=k} \prod_{i \in T} \delta_i \leq D_\delta^k$ and sum $\sum_{k \geq 3} D_\delta^k / (2^k n_0^{k+1}) = O(D_\delta^3 / n_0^4)$. Dividing by $W(u)$ gives $O(H/n_0)$. The total tail is at most $2^q E_0$, exponentially negligible relative to $W(u)$ because $q \log(n_0 + D_\delta) = o(n_0)$. □

Splitting large increments among repeated columns. The physical atom theorem requires each individual rail increment to be at most its baseline R. A large effective increment H can nevertheless be realized using repeated copies of one logical column. For even integers H, R, write H as a sum of $\lceil H/R \rceil$ positive even integers, each at most R. Assign those increments to that many identical columns. Every retained transversal chooses all their long rails together, so their scalar degree contribution is exactly $H(a \cdot s)$. Thus before the scalar Walsh analysis these copies may be collapsed into one effective increment, while the physical atom theorem keeps them separate. Repeated column directions do not change the Cayley independent-set condition used for observability.

The construction uses $R = \Theta(\log^2 m)$ and $H = \Theta(m \log m)$. It has $O(m)$ ordinary physical columns and $O(H/R) = O(m/\log m)$ additional copies in each distinguished group. Hence the total pair count is $p = \Theta(m)$ and $n_0 = pR + 1 = \Theta(m \log^2 m)$. On the auxiliary cube there are $q = O(m)$ effective columns, $S = O(m)$, and $D_\delta = O(m \log m) = o(n_0)$; all hypotheses above hold. After restriction to the logical column code, higher-order coefficients are pushed forward as sums. Negative singleton contributions occur only at individual columns. Neither the distinguished pair u nor any live candidate is such a column, so the required dominant coefficient and sign conditions are preserved.

7 A triangular path gives exponentially many distinct costs

We construct a column code with $\exp(\Omega(m))$ pointwise ordered scalar enumerators on m logical coordinates. All path increments belong to $\{2, 4, 6\}$. The argument uses the analytic cost expansion and an exact triangular transfer matrix, and requires no arithmetic independence assumption about optimized tanh values.

Lemma 15 (A positive inverse Laplace kernel near zero)**.** *Suppose a real-on-the-real-axis function w is holomorphic on a right half-plane and satisfies*

$$w(z) = Az + B + C/z + E(z), \qquad C > 0, \qquad |E(z)| \le K/|z|^2$$

uniformly there. There is a real continuous function ρ of at most exponential growth such that, for sufficiently large real n,

$$w(n) = An + B + \int_0^\infty e^{-nt}\rho(t)\,dt.$$

Moreover $\rho(t) \ge C/2$ for all sufficiently small positive t.

Proof. For an admissible vertical line $\Re z = c$, put

$$r(t) = \frac{1}{2\pi}\int_{\mathbb{R}} E(c+iy)e^{(c+iy)t}\,dy.$$

The integral is absolutely convergent. Closing rectangles proves independence of c for $t > 0$. Fubini followed by Cauchy's formula gives $E(z) = \int_0^\infty e^{-zt}r(t)\,dt$ for $\Re z > c$. Conjugation symmetry makes r real, and

$$|r(t)| \le \frac{Ke^{ct}}{2c}.$$

For small t choose $c = 1/t$, obtaining $|r(t)| \le eKt/2$. Thus $\rho = C + r$ is positive near zero. A fixed admissible c gives the claimed exponential-growth bound and continuity follows by dominated convergence. □

The interpolation in Theorem 7 satisfies this lemma after restricting to a sufficiently far-right half-plane.

7.1 An exactly triangular path transfer

Take a path on $V = 2k + 1$ vertices, where $k \ge 2$. Give every loop increment four, and give its successive edges increments $6, 2, 6, 2, \ldots, 6, 2$. For $t > 0$ put

$$a = \tanh(2t), \qquad e_j = \tanh(\delta_j t/2), \qquad M(a,e) = \begin{pmatrix} 1 & ae \\ a & e \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

If B is the path incidence matrix, the syndrome enumerator is

$$Z_v(t) = \sum_{y\in\mathbb{F}_2^{2k}} \left(\prod_{j=1}^{2k} e_j^{y_j}\right) a^{|v+By|}.$$

With $c_0 = (1, a)^T$ and $c_1 = (a, 1)^T$, successive edge summation gives

$$\begin{pmatrix} Z_v \\ Z_{v+e_1} \end{pmatrix} = J^{v_1}M(a,e_1)\cdots J^{v_{2k}}M(a,e_{2k})c_{v_{2k+1}}. \tag{27}$$

The first coordinate here is the scalar enumerator whose order is needed below.

Let

$$\mathsf{H} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \mathsf{D} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus $\mathsf{H}^{-1} = \mathsf{H}/2$ and $\mathsf{H}J\mathsf{H}^{-1} = \mathsf{D}$. Direct calculation gives

$$\mathsf{H}M(a,e)\mathsf{H}^{-1} = \alpha \begin{pmatrix} 1 & x \\ yx & y \end{pmatrix}, \qquad \alpha = \frac{(1+a)(1+e)}{2}, \quad x = \frac{1-e}{1+e}, \quad y = \frac{1-a}{1+a}. \tag{28}$$

Here $y = e^{-4t}$ and $x = e^{-\delta_j t}$.

Fix every even-position syndrome bit to one. Write $x = e^{-2t}$. The two steps of each block, with their positive factors α temporarily omitted, multiply to

$$\begin{pmatrix} 1 & x^3 \\ x^5 & x^2 \end{pmatrix} \mathsf{D} \begin{pmatrix} 1 & x \\ x^3 & x^2 \end{pmatrix} = \begin{pmatrix} 1-x^6 & x(1-x^4) \\ 0 & -x^4(1-x^2) \end{pmatrix}. \tag{29}$$

Consequently, if the preceding odd-position bit is η_i, the complete block equals a common positive scalar $L(t)$ times

$$T_i(t) = \begin{pmatrix} 1 & b(t) \\ 0 & \epsilon_i r(t) \end{pmatrix}, \qquad \epsilon_i = (-1)^{\eta_i + 1},$$

where

$$b(t) = \frac{x(1+x^2)}{1+x^2+x^4} > 0, \qquad r(t) = \frac{x^4}{1+x^2+x^4} \in (0, 1/3), \tag{30}$$

and

$$L(t) = \alpha_6(t)\alpha_2(t)(1-x^6) > 0.$$

The continuous limits at zero are $b(0) = 2/3$, $r(0) = 1/3$, and $L(t) = 3t + O(t^2)$.

Fix the terminal syndrome bit to zero. Let the k odd-position bits $\eta_1, \ldots, \eta_k$ vary subject to one prescribed total parity. Put

$$d_0 = 1, \qquad d_j = \prod_{i=k-j+1}^{k} \epsilon_i \quad (1 \le j \le k-1), \qquad E = \prod_{i=1}^{k} \epsilon_i.$$

The sign E is common to the entire family, while $(d_1, \ldots, d_{k-1})$ ranges freely over $\{-1, 1\}^{k-1}$. Multiplying the triangular matrices, then using $\mathsf{H}c_0 = (1+a, 1-a)^T$ and returning to the original first coordinate in (27), gives the exact scalar formula

$$Z_\eta(t) = \frac{L(t)^k}{2} \left[(1+a) + (1-a)b(t) \sum_{j=0}^{k-1} d_j r(t)^j + (1-a)E r(t)^k \right]. \tag{31}$$

The fixed total parity is what makes the final boundary term common.

Lemma 16 (Pointwise strict scalar order)**.** *The 2^{k-1} scalar enumerators in (31) have the same strict lexicographic order for every $t > 0$. For one fixed $t_0 > 0$ and an absolute C_1, every distinct pair satisfies*

$$|Z_\eta(t) - Z_{\eta'}(t)| \ge e^{-C_1 k} t^k \qquad (0 < t \le t_0). \tag{32}$$

Proof. If $j \geq 1$ is the first differing digit and $d_j = 1$, $d'_j = -1$, then $0 < r \leq 1/3$ implies

$$\sum_l (d_l - d'_l) r^l \geq 2r^j - 2\sum_{l>j} r^l \geq r^j > 0.$$

All factors multiplying this sum in (31) are positive. This proves the fixed pointwise order. On a sufficiently small fixed interval, b, $1-a$, and r are bounded below by positive absolute constants and $L(t) \geq ct$. Since $j \leq k-1$, these bounds give (32). □

7.2 Embedding into rectangle costs

Add $P = 2V + 1 = 4k + 3$ independent prefix coordinates and two distinguished coordinates. The logical dimension is

$$m = V + P + 2 = 6k + 6.$$

Use one effective loop column for every logical coordinate and all $V - 1 = 2k$ path-edge columns. The number of effective columns is $\ell = m + V - 1 = 8k + 6$. Put

$$H = 2\lceil m \log(m+2) \rceil, \qquad R = 2\lceil K \log^2(m+2) \rceil,$$

where K is a sufficiently large fixed constant. Give the distinguished loops effective increment H, every other loop increment four, and the path edges alternating increments six and two. Realize each distinguished loop by $g = \lceil H/R \rceil$ repeated physical columns, whose positive even increments sum to H and individually are at most R. Thus the physical pair count is

$$p = \ell - 2 + 2g = \Theta(m), \qquad n_0 = pR + 1 = \Theta(m \log^2 m).$$

The shared carrier and common baseline R give

$$n_s = n_0 + H(s_1 + s_2) + \sum_{j \text{ other}} \delta_j (a_j \cdot s), \qquad n_0 = pR + 1.$$

The bits in this degree formula are interpreted as zero or one. Splitting a distinguished column into repeated copies does not change its displayed effective degree contribution. Every physical increment is at most R, so the exact atom theorem applies for large enough K. Also $H/n_0 = O(1/\log m)$.

Let $u = e_1 + e_2$. Each candidate v_η has both distinguished bits and every prefix bit one. Its path syndrome has even-position bits one, terminal bit zero, and odd-position bits η of the prescribed total parity. Put

$$W(v) = 2^{-m} \sum_s (-1)^{s \cdot v} w(n_s), \qquad \nu_\eta = W(v_\eta) - W(v_\eta + u).$$

Neither candidate nor its translate by u is zero or a column, so the affine part $An + B$ has no coefficient there. Factoring the Laplace integrand gives

$$\nu_\eta = -\int_0^\infty e^{-(n_0+H)t} \rho(t) A_{\text{prefix}}(t) A_{\text{path}}(t) Z_\eta(t)\, dt. \tag{33}$$

Here $A_{\text{prefix}} = [(1 - e^{-4t})/2]^P$, and A_{path} is the product of the common positive Fourier factors $(1 + e^{-\delta_j t})/2$ for path loops and edges. The two distinguished factors give exactly

$$[(1 - e^{-Ht})/2]^2 - [(1 + e^{-Ht})/2]^2 = -e^{-Ht}.$$

Theorem 17 (Exponentially many ordered tanh rectangle costs)**.** *Assume $H \le C_2 n_0$ for a fixed constant C_2 and*

$$\frac{n_0}{m \log n_0} \longrightarrow \infty. \tag{34}$$

For all sufficiently large k, all 2^{k-1} costs in (33) are negative and pairwise distinct in a fixed strict order. Thus there are $\exp(\Omega(m))$ competing cost levels.

Proof. The sign of $Z_\eta - Z_{\eta'}$ is fixed at every positive time. On $t \in [1/n_0, 2/n_0]$, the kernel satisfies $\rho \ge C/2$, $A_{\text{prefix}} \ge t^P$, and A_{path} is bounded below by a positive absolute constant. The last assertion follows because its logarithm is $O(mt)$ and all path increments are at most six. Also $e^{-(n_0+H)t}$ is bounded below. Hence the positive near-zero contribution to a cost difference has magnitude at least

$$e^{-C_3 k} n_0^{-(P+k+1)} = \exp[-O(m \log n_0)]. \tag{35}$$

Choose fixed $t_0 > 0$ with $\rho \ge C/2$ on $(0, t_0]$. Every Fourier coefficient of $\exp[-t \sum_j \delta_j (a_j \cdot s)]$ is nonnegative and at most one, because its factor expansion has nonnegative coefficients with total sum one. Thus a difference of two complete coefficient integrands is at most two in absolute value after the common $e^{-n_0 t}$ factor is removed. The exponential-growth bound on ρ bounds the adverse tail $t \ge t_0$ by $\exp[-\Omega(n_0)]$. By (34), this is smaller than (35). The integral therefore preserves the fixed scalar order, with its overall negative sign.

The same argument without subtraction proves every $\nu_\eta < 0$: for a direct uniform lower bound, the all-loop term in Z_η is $a^{|v_{\text{path}}|} \ge t^{2k+1}$ on the small interval. The same lower bound with the two positive distinguished factors proves $W(v_\eta) > 0$. Consequently $W(v_\eta + u) > W(v_\eta) \ge 0$ throughout the affine family. □

7.3 Calibration and resource compatibility

The family of all candidates is itself an affine coset U: both distinguished bits and all prefix bits are fixed to one, even path bits are fixed to one, the terminal path bit is fixed to zero, and the free odd path bits have one prescribed total parity. A loop at a fixed bit moves outside U; a loop at a free bit changes its prescribed parity. Every path edge changes a fixed even-position bit, including the final edge. Thus $U \cup (U + u) \cup \{u\}$ is independent in the Cayley graph of all physical columns. The prefix separates u from the two cosets, and no column equals u. Repeated distinguished columns preserve this property.

The common-dual and physical-rounding arguments therefore apply unchanged once their dominant-Walsh inequalities hold. The P compulsory prefix factors give the uniform tail estimate

$$|\nu_v| \le O(2^P P!/n_0^{P+1}) + \exp[-\Omega(n_0)], \qquad v \in U.$$

When $n_0/m \to \infty$, this absorbs 2^m times every fixed polynomial stability and rounding factor under (34).

For the displayed parameter choice, the effective increment cube has $q = \ell = O(m)$ coordinates, total ordinary increment $S = O(m)$, and total increment $D_\delta = 2H + S = O(m \log m) = o(n_0)$. In particular $q \log(n_0 + D_\delta) = o(n_0)$. The inverse-Laplace Walsh estimates therefore apply, with relative off-pair mass

$$O(S/H + H/n_0) = O(1/\log m).$$

Higher-order coefficients push forward as sums under the column map; negative singleton contributions occur only at individual columns. Neither u nor a live candidate is a column, so the common-dual dominance and sign hypotheses hold for all large m.

The physical input dimension is

$$d_m = 1 + 2pR + 2H + S = \Theta(m \log^2 m).$$

Every target parity has degree $O(m \log^2 m)$. The same threshold-step tanh teacher consequently has width $C_m = O(m^2 \log^2 m)$, coefficients bounded by that displayed width, and uniform margin one. The designated mistake count is

$$T_m = 2^{k-1} = 2^{m/6-2} = \exp\left(\Omega\left(\frac{\sqrt{C_m}}{\log C_m}\right)\right).$$

The target's bounded-fan-in Boolean circuit has $O(d_m)$ gates; repeated columns merely contribute additional rail parities to the same two distinguished logical bits.

8 Completion of the online theorem

The path construction of Section 7 uses even rails and one shared carrier. Its supports have odd degrees $n_s = \Theta(m(\log m)^2)$. The sharp global code-atom theorem applies to every affine neuron, and its support bound is polynomial in these parameters. Appendix A therefore classifies every minimum-cost completion after every calibrated history. The compulsory prefix factors make the remaining Fourier costs small enough to absorb all polynomial stability and rounding constants. The strict ordering in Theorem 17 supplies $2^{k-1} = \exp(\Omega(m))$ designated rounds. Appendix B proves the conclusion for the actual Gaussian posterior at all sufficiently large fixed widths.

It remains to verify the teacher resources. The target is the sign of

$$b(x)\left(\sum_{i=1}^{m}(-1)^{z_i(x)} - (m-1)\right) = \sum_{i=1}^{m}\chi_{A_{e_i}}(x) - (m-1)\chi_{A_0}(x),$$

a score of margin one. Each parity has degree $O(m(\log m)^2)$. On the attainable values of the coordinate sum, parity is represented by an alternating sum of threshold steps. Replace each step at a midpoint by a tanh transition of gain $O(\log(mn_{\max}))$. Each attainable value is at distance at least one from every midpoint, so the uniform error can be made less than $1/(8m)$ per parity using $O(n_{\max})$ neurons. The combined score has margin greater than $1/2$; doubling its readout gives margin one. Its width is $O(mn_{\max}) = O(m^2(\log m)^2)$, and incoming weights, biases, and readouts are bounded by that width after increasing its fixed constant. For the Boolean circuit, compute each rail parity and the base parity once. Forming the logical bits uses only $O(\ell)$ further XOR gates, since every path column, including a repeated distinguished column, has at most two nonzero entries. An AND of the m logical zero tests and a final sign multiplication then give the target. The total bounded-fan-in gate count is $O(d_m)$. This proves Theorem 1.

A Control of physical network competitors

The sharp code-atom theorem and its quantitative support bound control all physical network competitors, including those whose outputs have Fourier components outside the logical code. We now prove that every minimum-cost feasible score is the cheapest live rectangle.

Here are the physical definitions, allowing the column code of Section 7. Take ℓ disjoint rail pairs B_j^0, B_j^1 and nonzero columns $a_j \in \mathbb{F}_2^m$ spanning the logical space, together with an independent shared carrier $x_\star$. For a uniformly random physical input put

$$y_j^a = \chi_{B_j^a}(x), \qquad b = x_\star \prod_j y_j^0, \qquad (-1)^{\zeta_j} = y_j^0 y_j^1, \qquad z = \sum_j a_j \zeta_j.$$

The logical vector z is uniform on $G = \mathbb{F}_2^m$. The code space $\mathcal{C}$ is the span of physical characters $\chi_{A_s} = b(-1)^{s\cdot z}$, where $A_s = \{\star\} \cup \bigcup_j B_j^{a_j \cdot s}$. The odd downward Fourier space $\mathcal{D}$ is spanned by odd-degree characters with support contained in some A_s. All physical rail sizes are even. The independent-rail code is the special case $a_j = e_j$. All projections and norms are in the uniform physical L^2 space. In particular,

$$P_{\mathcal{C}} F = b\, g(z), \qquad g(z) = \mathbb{E}[bF \mid z].$$

The target is $f(x) = b(x)$ when $z = 0$, and $f(x) = -b(x)$ otherwise. Each revealed logical fiber means that all its physical inputs have been labeled. Closed global margins mean $|F(x)| \geq 1/2$ everywhere; correct labels on revealed fibers mean $f(x)F(x) \geq 1/2$ there.

Let $M = 2^m$, $m \geq 6$. Write $w_s \geq w_0 > 0$ for code weights and $W(z) = M^{-1} \sum_s w_s (-1)^{s\cdot z}$. Set $u = e_1 + e_2$. Take an affine coset $U = v_0 + H$ with $K_U = |U| \leq M/32$, where $u \notin H$ and $v_0 \notin H + \langle u \rangle$. Put $H' = H + \langle u \rangle$, $U' = U + \langle u \rangle$, and $E' = H'^{\perp} \setminus (H' + \langle v_0 \rangle)^{\perp}$. Assume $U' \cup \{u\}$ is independent in the Cayley graph whose edges add one of the columns a_j. In the independent-rail case, one can fix its first $q \geq 4$ bits and require odd total parity. The path coset in Section 7 also satisfies these conditions. Now $|E'| = M/(4K_U) \geq 8$, every $s \in E'$ has $s \cdot u = 0$, and

$$\sum_{s \in E'} (-1)^{s\cdot z} = |E'|(\mathbf{1}_{H'}(z) - \mathbf{1}_{U'}(z)).$$

For every nonzero coset c of $\langle u \rangle$, put

$$k_c = \max\{0, W(z), W(z+u)\} \quad (z \in c), \qquad K_* = \sum_{c \neq 0} k_c, \qquad w_* = W(u).$$

Assume

$$0 < w_* \leq w_0/16, \qquad K_* \leq w_*/4, \qquad M w_* \geq 16 w_0, \qquad W(v+u) > W(v) \geq 0 \quad (v \in U). \tag{36}$$

A.1 The common dual

Put $L = M/2$ and define

$$h = \frac{(L-1)(w_* - 4w_0/M) + K_*}{L-2}, \qquad b_* = \frac{M(h + K_*)}{L-1}, \qquad a_* = 2w_0 - b_*.$$

These satisfy $3w_*/4 \leq h \leq 5w_*/4$, $b_* \leq 4w_* \leq w_0/4$, and $7w_0/4 \leq a_* \leq 2w_0$. Set

$$r_c = \frac{(L-1)k_c}{Lh + K_*}, \qquad R_* = \sum_{c \neq 0} r_c \leq 1/3.$$

On $u^{\perp}$ define the strictly positive probability distribution

$$P(s) = \frac{1}{L}\left(1 + \sum_{c \neq 0} r_c (-1)^{s \cdot z_c}\right).$$

Its mass at zero is

$$p = P(0) = \frac{h + K_*}{Lh + K_*}, \qquad 1 - p = \frac{(L-1)h}{Lh + K_*}.$$

Let Q be P conditioned on $s \neq 0$, extended by zero outside $u^\perp \setminus \{0\}$. Define a logical dual by

$$\widehat{\lambda}(s) = w_s - 2w_0 \mathbf{1}_{s=0} - MhQ(s) - \frac{a_*}{|E'|}\mathbf{1}_{s\in E'}. \tag{37}$$

For every s, $|\widehat{\lambda}(s)| \le w_s$. Indeed its value at zero is $-w_0$, and it equals w_s if $s \cdot u = 1$. Otherwise $MhQ(s) \le Mhp/(1-p) = b_*$, so the total subtraction is at most $b_* + a_*/|E'| \le 2w_0 \le 2w_s$.

For $z \notin \{0, u\}$, Fourier orthogonality gives

$$\mathbb{E}_Q(-1)^{s\cdot z} = \frac{k_{[z]}}{h} - \frac{h + K_*}{(L-1)h}.$$

Consequently

$$\lambda(z) = M(W(z) - k_{[z]}) - a_* - a_*\mathbf{1}_{H'}(z) + a_*\mathbf{1}_{U'}(z) \quad (z \neq 0, u). \tag{38}$$

The defining equation for h also gives

$$\lambda(u) = 0, \qquad \lambda(0) = M(W(0) - w_*) \ge a_*.$$

In particular,

$$\lambda(v) = M\nu_v < 0, \qquad \lambda(v+u) = 0, \qquad \nu_v := W(v) - W(v+u) \quad (v \in U).$$

On the observed logical region $\mathcal{O} = G \setminus (U' \cup \{u\})$, its sign is positive at zero and negative elsewhere, with magnitude at least a_*.

There is also a useful strict coefficient slack. If $s \neq 0$ and $s \cdot u = 0$, then

$$\min\{w_s - \widehat{\lambda}(s), w_s + \widehat{\lambda}(s)\} \ge w_*. \tag{39}$$

For the first term use $MhQ(s) \ge 2h(1-R_*)/(1-p) \ge w_*$. For the second use $2w_s - b_* - a_*/|E'| \ge a_*(1 - 1/|E'|) > w_*$.

A.2 Observability outside a logical independent set

Lemma 18. *If $J \subseteq G$ is independent in the Cayley graph generated by the columns a_j, then every $A \in \mathcal{D} \cap \mathcal{C}^\perp$ satisfies*

$$\|A\|_2^2 \le 2\mathbb{E}[\mathbf{1}_{z\notin J} A^2].$$

Proof. Every Fourier support T contributing to A is an odd-degree proper subset of a carrier-containing transversal. It cannot occupy a full rail at every pair: with the carrier it would be a retained code character, while without the carrier it would have even degree. Assign it to the first rail pair i at which its restriction is neither complete rail. Write $A = \sum_i A_i$ for this partition.

Conditioned on a logical fiber, two physical characters have nonzero inner product only when their symmetric difference is a union of full rail pairs with indicator vector $(a_j \cdot s)_j$ for some s. A symmetric difference containing the carrier has zero conditional expectation, since the carrier is uniform and independent on each logical fiber. If one support is a proper subset of a rail at pair i, its complement in that pair cannot be contained in a single rail. Thus conditional covariance with another downward character requires the same restriction at pair i. It follows that A_i, A_j are conditionally orthogonal whenever $i \neq j$, and that $\ell_i(z) := \mathbb{E}[A_i^2 \mid z]$ is invariant under $z \mapsto z + a_i$. Indeed every surviving logical Fourier character has $a_i \cdot s = 0$.

An independent set contains at most one endpoint of every edge that adds a_i. Therefore $\mathbb{E}[\mathbf{1}_{z\in J}\ell_i(z)] \le \|A_i\|_2^2/2$. Sum over i, using conditional orthogonality, to obtain the assertion. □

By the assumed Cayley independent-set condition, Lemma 18 applies to $\mathcal{O}$.

A.3 Physical variance control

Suppose a score F and a nonnegative cost $B \le B_{\mathrm{cap}}$ satisfy

$$Q(F) := \sum_s w_s |\widehat{F}(A_s)| \le B, \qquad \|F\|_\infty \le H_{\mathrm{cap}}, \qquad \|F - P_{\mathcal{D}}F\|_2^2 \le S_{\mathrm{at}} B D,$$

where $D = B - \langle b\lambda, F\rangle \ge 0$ and b is the physical base character. These bounds follow from the sharp atom theorem and its support stability estimate, namely

$$Q(\phi_a) \le \|a\|_2, \qquad \|\phi_a - P_{\mathcal{D}}\phi_a\|_2^2 \le S_{\mathrm{at}}\|a\|_2(\|a\|_2 - Q(\phi_a)), \quad \phi_a(x) = \tanh(a\cdot\bar{x}),$$

for a representation $F = \sum_j c_j \phi_{a_j}$ of cost $B = \sum_j |c_j|\|a_j\|_2$. Indeed the dual inequality gives $|\langle b\lambda, \phi_a\rangle| \le Q(\phi_a)$, and Cauchy–Schwarz bounds the squared sum of residuals by $S_{\mathrm{at}} B D$. The elementary bound $|\phi_a(x)| \le \sqrt{d+1}\|a\|_2$ also gives $H_{\mathrm{cap}} = \sqrt{d+1}B_{\mathrm{cap}}$. The atom theorem supplies a polynomial value of S_{at}. Assume closed full-cube margins and correct labels outside U, with logical target positive only at zero. Extra fully revealed fibers in U have the negative label. Write $P_{\mathcal{C}}F = bg$ and

$$S = \mathbb{E}[\mathbf{1}_{z\in\mathcal{O}}|\lambda(z)|(fF - 1/2)] \ge 0.$$

On each observed fiber, the variable $Y = fF$ is at least $1/2$. Thus $\mathbb{E}[|Y - \mathbb{E}Y| \mid z] \le 2(\mathbb{E}[Y \mid z] - 1/2)$. Since $|\lambda| \ge a_*$ there, this proves $\mathbb{E}[\mathbf{1}_{\mathcal{O}}|F - P_{\mathcal{C}}F|] \le 2S/a_*$. Applying independent-set observability to $P_{\mathcal{D}}F - P_{\mathcal{C}}F$ therefore yields

$$L_F := \|F - P_{\mathcal{C}}F\|_2^2 \le C_{\mathrm{phys}}(D + S), \qquad C_{\mathrm{phys}} = \max\{16H_{\mathrm{cap}}/a_*,\ 5S_{\mathrm{at}}B_{\mathrm{cap}}\}. \tag{40}$$

To see the constants, write $R = F - P_{\mathcal{C}}F = A + E$, where $A = P_{\mathcal{D}}F - P_{\mathcal{C}}F$ and $E = F - P_{\mathcal{D}}F$. The preceding conditional bound and $|R| \le 2H_{\mathrm{cap}}$ give $\mathbb{E}[\mathbf{1}_{\mathcal{O}}R^2] \le 4H_{\mathrm{cap}}S/a_*$. Observability and $(R-E)^2 \le 2R^2 + 2E^2$ imply $\|A\|_2^2 \le 4\mathbb{E}[\mathbf{1}_{\mathcal{O}}R^2] + 4\|E\|_2^2$. Adding $\|E\|_2^2$ proves (40).

A.4 Linear rounding to zero or a live rectangle

Put $p_v = g(v) + 1/2$ for $v \in U$ and $\delta = D/w_* + S/a_*$. By (39), the nonconstant u-invariant part r of g satisfies $\|r\|_\infty \le D/w_*$. The region $\mathcal{O}_0 = G \setminus (U' \cup \{0, u\})$ is invariant under u, has density at least $1/2$, and consists of known negative fibers. Averaging $g + 1/2$ there eliminates the u-anti-invariant part. Consequently

$$|\widehat{g}(0) + 1/2| \le D/w_* + 2S/a_*, \qquad \sup_z |g(z) + g(z+u) + 1| \le 4\delta.$$

Each partner $v + u$ is a known negative fiber, so $p_v \ge -4\delta$. In particular the total negative mass of p is at most $M\delta$.

Compare g with

$$g_{\mathrm{ref}} = -1/2 + \mathbf{1}_0 - \mathbf{1}_u + \sum_{v\in U} p_v(\mathbf{1}_v - \mathbf{1}_{v+u}).$$

The logical ℓ^1 norm of their difference is at most

$$2MS/a_* + 4(K_U + 1)\delta \le 3M\delta.$$

Here the observed fibers contribute at most MS/a_*; the fiber u contributes at most the anchor excess plus 4δ; and the K_U partner fibers contribute at most $4K_U\delta$.

There are characters s with $s \cdot u = 1$ and $(-1)^{s \cdot v} = -1$ for every $v \in U$: prescribe the two independent linear conditions $s \cdot u = 1$ and $s \cdot v_0 = 1$ within $H^{\perp}$. For any such s, the dual is saturated on its positive side, so $\widehat{g}(s) \geq -D/(2w_s)$. Since

$$\widehat{g}_{\mathrm{ref}}(s) = \frac{2}{M}\left(1 - \sum_{v \in U} p_v\right),$$

we obtain

$$\sum_{v \in U} p_v \leq 1 + \frac{MD}{4w_0} + \frac{3}{2}M\delta \leq 1 + 2M\delta.$$

Thus the total positive mass is at most $1 + 3M\delta$.

On a fiber with $0 < p_v < 1$, the mean $g(v) = p_v - 1/2$ lies between the two allowed output intervals. A variable $Y = bF$ with $|Y| \geq 1/2$ and mean $g(v)$ has variance at least $1/4 - g(v)^2 = p_v(1 - p_v)$. Consequently

$$V_{\mathrm{frac}} := \sum_{0 < p_v < 1} p_v(1 - p_v) \leq ML_F.$$

Round each p_v to its nearest $b_v \in \{0, 1\}$. Negative coordinates contribute at most $M\delta$ to the rounding distance. Coordinates above one contribute at most $3M\delta$, because total positive mass is at most $1 + 3M\delta$. The fractional coordinates contribute at most $2V_{\mathrm{frac}}$. Hence $A := \|p - b\|_1 \leq 4M\delta + 2V_{\mathrm{frac}}$. If $b = 0$, choose the zero vector. Otherwise choose one coordinate with $b_v = 1$; it is live because removed fibers have $p_v \leq 0$. The distance from b to this point mass is at most $3M\delta + A$. Therefore

$$\mathrm{dist}_1(p, \{0\} \cup \{e_v : v \text{ live}\}) \leq 11M\delta + 4V_{\mathrm{frac}} \leq B_{\mathrm{round}}(D + S), \tag{41}$$

where

$$B_{\mathrm{round}} = M(11/w_* + 11/a_* + 4C_{\mathrm{phys}}).$$

A.5 Absorption and classification of every minimum

The baseline score $g_0 = -1/2 + \mathbf{1}_0 - \mathbf{1}_u$ and rectangle scores $g_v = g_0 + \mathbf{1}_v - \mathbf{1}_{v+u}$ saturate the common dual. Their weighted Fourier costs are

$$C_0 = w_0/2 + W(0) - W(u), \qquad C_v = C_0 + \nu_v.$$

The exact calibrated decomposition is

$$B = C_0 + D + S + \sum_{v \in U} \nu_v p_v.$$

No omitted-fiber term remains: $\lambda(u) = 0$ and $\lambda(v + u) = 0$ for all $v \in U$. Let $W_{\mathrm{eff}} = \max_{v \in U} |\nu_v|$. If $W_{\mathrm{eff}} B_{\mathrm{round}} < 1$, rounding gives

$$B \geq C_{\mathrm{choice}} + (1 - W_{\mathrm{eff}} B_{\mathrm{round}})(D + S),$$

where the choice is zero or a live point mass. The zero choice has cost C_0, strictly greater than every live star because $\nu_v < 0$. Consequently any minimum-cost completion is a cheapest live rectangle. If its effective costs are pairwise distinct, the score is unique: equality forces $D = S = 0$, then $L_F = 0$, exact anti-invariance around $-1/2$, and the rounded point mass determine every logical and physical value.

Lemma 14 and the scalar construction make (36) hold for large m. The atom theorem gives $S_{\mathrm{at}} = 2\max(2, 96\sqrt{6}\, W^2) = O(n_0^2)$, and one may take $B_{\mathrm{cap}} = O(n_0)$. Every factor in B_{round} except

2^m is therefore polynomial in m. The compulsory prefix gives the required small Fourier costs. On $(0, t_0]$, the Laplace kernel is bounded and $A_{\text{prefix}}(t) \le (2t)^P$, while the combined remaining coefficient factors are at most one. Therefore

$$|\nu_v| \le O(2^P P!/n_0^{P+1}) + \exp(-\Omega(n_0)),$$

uniformly over the affine live envelope. Now $P = \Theta(m)$ and $n_0 = \Theta(m \log^2 m)$, so this bound is $\exp[-\Omega(m \log\log m)]$ and absorbs 2^m times any fixed polynomial. The explicit polynomial atom stability and the polylogarithmic rail baseline permit this resource choice.

This proves the calibration statement for the scalar construction. Section 7 supplies the distinct effective costs, and Appendix B transfers the score-space cost gaps to the actual fixed-width Gaussian posterior.

B Finite-width energy and the actual posterior

The preceding sharp atomic and calibration theorems identify every minimum-cost score. We now transfer their strict cost gaps to every sufficiently large fixed width and compare the actual Gaussian posterior integrals. The Gaussian variances are those in Section 1.

Lemma 19 (Integer allocation of a finite atom representation)**.** *Let $F = \sum_{j=1}^{J} c_j \tanh(a_j \cdot \bar{x})$ be a finite representation with nonzero a_j, c_j. Put*

$$B = \sum_j |c_j| \|a_j\|_2, \qquad p_j = \frac{|c_j|}{\sqrt{d}\,\|a_j\|_2}, \qquad p = \sum_j p_j.$$

If $p < 1$, then for every sufficiently large integer N there is an exact width-N representation of F with Gaussian energy tending to $2\sqrt{d}\,B$ as $N \to \infty$.

Proof. Choose $r_j = \lceil N p_j \rceil$. For $N(1-p) \ge J$, $\sum_j r_j \le N$. Use r_j copies of the incoming vector a_j, each with readout c_j/r_j, and pad with zero neurons. The output is exactly F. Direct calculation gives

$$K_N - 2\sqrt{d}\,B = \sum_j \frac{d\|a_j\|_2^2}{N} \frac{(r_j - N p_j)^2}{r_j} \le \frac{d}{N} \sum_j \|a_j\|_2^2.$$

The right side tends to zero. The same statement applies after any fixed small multiplicative increase of all readouts, provided the resulting allocation sum remains less than one. □

For the balanced parity-orbit representation of a code score $bg(z) = \sum_s \widehat{g}(s)\chi_{A_s}$, let q_n denote its positive parity correlation at the optimal scale a_n. The finite orbit identity represents each parity using incoming vectors of norm $a_n\sqrt{n}$ and total absolute readout $1/q_n$. Therefore

$$B = \sum_s w_{n_s} |\widehat{g}(s)|, \qquad p = \frac{1}{\sqrt{d}} \sum_s \frac{|\widehat{g}(s)|}{a_{n_s}\sqrt{n_s}\,q_{n_s}}.$$

The balanced scalar asymptotics imply $\inf_{n \ge n_*} a_n\sqrt{n}\,q_n > 0$. Every baseline or rectangle score in the construction has $\sum_s |\widehat{g}(s)| = 3/2$. Thus $p = O(d^{-1/2}) < 1/2$ for all sufficiently large construction parameters. Integer allocation applies uniformly to the finitely many rectangle witnesses needed for a fixed instance, and also to a small strict-margin dilation of each witness.

Here is the output-space gap needed for the Bayesian argument. Fix a history with at least one live point and pairwise distinct effective rectangle costs. Choose a polynomial cap B_{cap} strictly

above all rectangle costs. Consider all pairs (B, F) obeying the closed atomic score inequalities of Appendix A, the cap, $\|F\|_\infty \le \sqrt{d+1}B_{\mathrm{cap}}$, the closed global margins, and all revealed labels. This is a compact subset of a finite-dimensional space. The calibration proof says that its unique minimum-cost score is the cheapest live rectangle, of cost C_*. That score is wrong at its exceptional live fiber.

Intersect the compact set with the additional condition that the score is correct at the next chosen physical input in that fiber. This closed subset has minimum cost $C_{\mathrm{good}} > C_*$. If it is empty, set $C_{\mathrm{good}} = +\infty$. Actual feasible network representations of cost at most B_{cap} belong to the compact set by the atomic theorem and their full-cube and history margins. Representations above the cap have even larger energy because $K_N \ge 2\sqrt{d}\,B$. Consequently the energy of every feasible network correct at the designated input is at least

$$E_{\mathrm{good}} = 2\sqrt{d}\,\min(C_{\mathrm{good}}, B_{\mathrm{cap}}) > 2\sqrt{d}\,C_*.$$

Choose a small fixed dilation of the cheapest rectangle so that its margins are strict and its variation cost remains below $E_{\mathrm{good}}/(2\sqrt{d})$. Integer allocation gives, for every sufficiently large N, a strict feasible wrong network of energy strictly below E_{good}.

For completeness, fix one such N and write $P = N(d+2)$ for the parameter dimension. A small open ball around the strict feasible witness lies in the conditioned version space and has energy bounded by some $a < E_{\mathrm{good}}$. The correct region has energy at least E_{good}. Choose $\rho \in (0,1)$ so that $(1-\rho)E_{\mathrm{good}} > a$. Its unnormalized Gaussian integral is at most

$$e^{-(1-\rho)E_{\mathrm{good}}/(2\sigma^2)} \int_{\mathbb{R}^P} e^{-\rho K_N(\theta)/(2\sigma^2)}\, d\theta = C_{N,\rho}\sigma^P e^{-(1-\rho)E_{\mathrm{good}}/(2\sigma^2)},$$

whereas the version-space integral is at least the ball volume times $e^{-a/(2\sigma^2)}$. Their ratio tends to zero. Thus the actual conditioned Gaussian posterior predicts the wrong sign with probability tending to one as $\sigma \downarrow 0$ at this fixed width.

There are finitely many logical rounds in each fixed construction. Take the maximum of their width thresholds. The posterior conclusion then holds for every fixed width beyond that threshold and every round.

Pairwise distinctness of every candidate is stronger than necessary. If there are T distinct effective cost levels, first reveal every candidate fiber except one representative per level. At each subsequent round, take one physical input in the cheapest surviving representative fiber, then reveal the remainder of that fiber before continuing. All these labels are target labels, all inputs are distinct, and the same analysis applies at each designated prediction. This gives T designated mistakes with posterior probability tending to one at each round. Section 7 establishes exponentially many such levels in the logical dimension for balanced tanh costs on its path column code.

C Signed periodic responses and harmonic defects

This appendix proves the periodic-response and harmonic-defect estimates used in the global single-parity theorem.

For $a > 0$, define the paired series

$$T_a(b) = \tanh b + \sum_{j\ge 1} (-1)^j [\tanh(b+2aj) + \tanh(b-2aj)]. \tag{42}$$

It and every fixed derivative converge locally uniformly for $a > 0$. The function is odd and satisfies $T_a(b+2a) = -T_a(b)$.

Theorem 20 (The exact signed periodic maximum)**.** *For every $a > 0$,*

$$T_a'(b) > 0 \quad (-a < b < a), \qquad T_a'(a) = 0, \qquad T_a''(a) < 0.$$

Consequently

$$|T_a(b)| \le T_a(a) = F_0(a),$$

with equality exactly at $b \in (2\mathbb{Z}+1)a$. There is an absolute $c > 0$ such that, for $a \in [1,2]$ and every real b,

$$F_0(a) - |T_a(b)| \ge c \operatorname{dist}(b, (2\mathbb{Z}+1)a)^2. \tag{43}$$

Proof. Euler's product for cosh gives

$$\operatorname{sech}^2 x = \prod_{j\ge 0} (1 + x^2/\lambda_j)^{-2}, \qquad \lambda_j = ((2j+1)\pi/2)^2.$$

Let S_j be independent gamma variables of shape two and rate λ_j. Their sum S is finite and positive almost surely, because $\sum_j \mathbb{E}S_j < \infty$. Their Laplace transforms give

$$\operatorname{sech}^2 x = \mathbb{E}e^{-Sx^2}. \tag{44}$$

For fixed $s > 0$, the alternating Gaussian sum

$$G_{a,s}(b) = \sum_{j\in\mathbb{Z}} (-1)^j e^{-s(b+2aj)^2}$$

is, after multiplication by $\sqrt{s/\pi}$, the Dirichlet heat kernel on $(-a, a)$, started at zero, at time $1/(4s)$. The method-of-images formula gives this identity directly. Positivity of the heat kernel and the boundary point lemma imply

$$G_{a,s}(b) > 0 \ (|b| < a), \qquad G_{a,s}(a) = 0, \qquad G_{a,s}'(a) < 0.$$

On the other hand,

$$T_a'(b) = \sum_{j\in\mathbb{Z}} (-1)^j \operatorname{sech}^2(b + 2aj) = \mathbb{E}G_{a,S}(b).$$

Interchange of sum and expectation is justified by the finite sum of absolute expectations $\sum_j \operatorname{sech}^2(b + 2aj)$. At the boundary, differentiation is also justified: the absolute expectations of the differentiated terms sum to $\sum_j |(\operatorname{sech}^2)'((2j+1)a)| < \infty$. Thus $T_a''(a) = \mathbb{E}G_{a,S}'(a) < 0$.

Oddness and anti-periodicity prove the maximum statement. Substituting $b = a$ into (42) and telescoping the exponentially small tails of $\tanh x = 1 - 2/(1 + e^{2x})$ gives

$$T_a(a) = 1 - 4\sum_{j\ge 0} \frac{(-1)^j}{1 + e^{2a(2j+1)}} = F_0(a).$$

Finally, the deficit in (43) has a strictly positive quadratic coefficient at each maximum. Away from the maxima it is strictly positive. Compactness in $a \in [1,2]$ and in the normalized periodic phase proves the uniform bound. □

Theorem 21 (Stability of an arbitrary number of phase defects)**.** *Let $k \ge 1$, let $\eta \in \{0,1\}$ with $k + \eta$ odd, and let $b_1, \dots, b_k \in \mathbb{R}$. For $a \in [1,2]$, put*

$$H = \mathbb{E}_Y \left[\chi_{[k]}(Y) T_a \left(\eta a + \sum_i b_i Y_i \right) \right].$$

There is an absolute $c > 0$, independent of k, such that

$$F_0(a) - |H| \geq c \min\left\{\sum_i \operatorname{dist}(b_i, (2\mathbb{Z}+1)a)^2, 1\right\}. \tag{45}$$

The extremal equality $|H| = F_0(a)$ holds exactly when every b_i is an odd integer multiple of a.

Proof. Choose odd integers r_i with $d_i = b_i - r_i a \in [-a, a]$. Anti-periodicity cancels the parity character, giving, up to a single common sign,

$$H = \mathbb{E}T_a\left(a + \sum_i d_i Y_i\right).$$

Theorem 20, including its strict second derivative, implies the uniform pointwise inequalities

$$F_0(a) \mp T_a(a+z) \geq c_0[1 \mp \cos(\pi z/(2a))].$$

Indeed the ratios extend positively across the appropriate maxima and are positive and periodic elsewhere. Taking expectations, the cosine average is

$$p = \prod_i \cos(\pi d_i/(2a)) \in [0, 1].$$

Both signed deficits are therefore at least $c_0(1-p)$. Since $\cos t \leq e^{-t^2/2}$ on $[-\pi/2, \pi/2]$,

$$1 - p \geq 1 - \exp\left(-\frac{\pi^2}{8a^2}\sum_i d_i^2\right) \geq c_1 \min\left\{\sum_i d_i^2, 1\right\}.$$

This proves the claim and the equality classification. □

Lemma 22 (Restoring curvature under phase averaging)**.** *For $a \in [1, 2]$, write $F = F_0(a)$ and $U(z) = T_a(a+z)$. There are positive parameters A and $\mu < 3/4$, depending on a, such that*

$$U''(z) \leq -A(1-\mu)F + A(1-\mu/2)(F - U(z)). \tag{46}$$

In particular, for any real random variable Z,

$$\mathbb{E}U(Z) \geq \tfrac{4}{5}F \quad \Longrightarrow \quad \mathbb{E}U''(Z) \leq -\tfrac{1}{2}A(1-\mu)F < 0.$$

Proof. Let the Jacobi elliptic modulus k have nome $q = e^{-\pi^2/(2a)}$, and let K be its complete elliptic integral. The Fourier series for the Jacobi function sn gives

$$T_a(b) = F_0(a)\operatorname{sn}(Kb/a, k), \qquad F_0(a) = kK/a.$$

This follows by comparison with

$$T_a(b) = \frac{\pi}{a}\sum_{j\geq 0}\frac{\sin((2j+1)\pi b/(2a))}{\sinh((2j+1)\pi^2/(4a))}.$$

Set $A = (K/a)^2$ and $\mu = k^2$. The differential equation for sn is

$$U'' = -A(1+\mu)U + 2A\mu U^3/F^2.$$

For $x = U/F \in [-1, 1]$, subtracting the right side of (46) from this expression gives $-AF\mu(1 - x)(2x + 1)^2/2 \le 0$.

For completeness, $q < 17/200$ on the indicated gain interval. The theta series imply

$$\mu = \left(\frac{\vartheta_2(0, q)}{\vartheta_3(0, q)}\right)^4 \le \frac{16q[1 + q^2/(1 - q^4)]^4}{(1 + 2q)^4} < \frac{3}{4}.$$

The rational upper bound increases for $q \le 17/200$ and is less than $3/4$ at $17/200$. The bound on q follows from $\pi > 3141/1000$ and a positive Taylor lower bound for the exponential. Finally, $(1 - \mu)/(1 - \mu/2) \ge 2/5$, which proves the expectation assertion. □

C.1 Exact harmonic vectors

The following argument treats a growing number of odd harmonics, including harmonics of unbounded size.

Theorem 23 (Odd harmonics cannot improve the balanced maximum)**.** *For all sufficiently large odd n, every $a \in [1, 2]$, and every vector of positive odd integers $r_1, \dots, r_n$,*

$$\frac{\left|\mathbb{E}\chi_{[n]}(X) \tanh(a \sum_i r_i X_i)\right|}{a\sqrt{\sum_i r_i^2}} \le \frac{q_n(a_n)}{a_n\sqrt{n}}.$$

Equality holds exactly when every $r_i = 1$ and $a = a_n$. The same statement allows arbitrary signs of the coefficients.

Proof. Let k count the r_i different from one and put $\Delta = \sum_i (r_i^2 - 1) \ge 8k$. The case $k = 0$ is the balanced-gain theorem. Suppose first $1 \le k \le n/2$. The Fourier representation

$$\tanh x = \int_0^\infty \frac{\sin(\omega x)}{\sinh(\pi\omega/2)}, d\omega$$

can be folded onto $(0, \pi)$ after setting $\theta = a\omega$. Because every r_i and n are odd, the product of sines changes sign between consecutive half-periods. The folded kernel is

$$K_a(\theta) = \sum_{j \ge 0} \frac{(-1)^j}{\sinh(\pi(\theta + j\pi)/(2a))} > 0.$$

The alternating-series test proves its strict positivity. For real $p > 0$, define

$$J_p(a) = \int_0^\pi \sin^p \theta \, K_a(\theta) \, d\theta.$$

For odd n, $q_n(a) = J_n(a)/a$. Dropping the absolute values of the k exceptional sine factors bounds the weighted coefficient by $J_{n-k}(a)/a$.

Uniformly for $a \in [1, 2]$, Laplace expansion at $\theta = \pi/2$ gives

$$J_p(a) = \sqrt{2\pi}\, K_a(\pi/2) p^{-1/2} \left[1 + \frac{b_a}{p} + O(p^{-2})\right], \qquad b_a = \frac{K_a''(\pi/2)}{2K_a(\pi/2)} - \frac{1}{4}. \tag{47}$$

To justify uniformity, expand $\log\cos x = -x^2/2 - x^4/12 + O(x^6)$ and $K_a(\pi/2 + x)$ on a fixed central interval. The remaining region is exponentially small. At the endpoint singularity, split

$\sin^p = \sin^{p/2}\sin^{p/2}$; one factor is exponentially small, and the other makes the inverse-linear kernel singularity uniformly integrable for $p \geq 2$. Consequently

$$\frac{J_{n-k}(a)}{J_n(a)} = \sqrt{\frac{n}{n-k}}\left[1 + O(k/n^2)\right] \quad (1 \leq k \leq n/2).$$

After dividing by the incoming norm and by the balanced ratio at gain a, the result is at most

$$\frac{n}{\sqrt{(n-k)(n+8k)}}[1 + O(k/n^2)] \leq 1 - \frac{k}{4n} < 1$$

for all sufficiently large n. The elementary first-factor inequality used here is $n/\sqrt{(n-k)(n+8k)} \leq 1 - k/(2n)$.

For $k \geq n/2$, use the universal bound $|\widehat{h}([n])| \leq \kappa_n$ for a monotone function $h : \{-1,1\}^n \to [-1,1]$. One proof averages over coordinate permutations and writes the resulting monotone sequence as a convex combination of Hamming-level thresholds and constants. The threshold coefficients are signed values $2^{1-n}\binom{n-1}{j}$, maximized at the center. Thus the normalized coefficient is at most $\kappa_n/\sqrt{5n}$. The balanced witness at gain one has ratio

$$\frac{q_n(1)}{\sqrt{n}} = \frac{\kappa_n}{\sqrt{n}}[F_0(1) + O(n^{-1})], \qquad F_0(1) \geq 1 - \frac{4}{1+e^2} > \frac{1}{2} > \frac{1}{\sqrt{5}}.$$

It is strictly better for large n. Coordinate sign changes affect only the overall sign of the parity coefficient. □

For a fixed number of fixed odd harmonics there is a more precise loss. If $\Delta = \sum_i (r_i^2 - 1)$ is fixed as $n \to \infty$, then uniformly for $a \in [1,2]$,

$$\frac{\left|\mathbb{E}\chi_{[n]} \tanh(a\sum_i r_i X_i)\right|}{a\sqrt{n+\Delta}} = \frac{q_n(a)}{a\sqrt{n}}\left[1 - \frac{\Delta}{n} + O(n^{-2})\right]. \tag{48}$$

Indeed

$$P_r(y) = (-1)^{(r-1)/2}\frac{\sin(r \arcsin\sqrt{y})}{\sqrt{y}}$$

is a polynomial with $P_r(1) = 1$ and $P_r'(1) = (r^2-1)/2$. Writing $\prod_i P_{r_i}(y) = \sum_j c_j y^j$, the Fourier integral expresses the coefficient, up to its fixed phase, as $\sum_j c_j q_{n+2j}(a)$. The balanced expansion gives $q_{n+2j}/q_n = 1 - j/n + O(n^{-2})$ for fixed j. Use $\sum_j c_j = 1$ and $\sum_j j c_j = \Delta/2$, then divide by $\sqrt{n+\Delta}$, to obtain (48).

C.2 A global maximum on every fixed-dimensional defect slice

Theorem 24 (A fixed number of bounded real defects)**.** *Fix an integer $k \geq 1$ and $B > 2$. For all sufficiently large odd n, among vectors with $n-k$ coordinates equal to $a \in [1,2]$ and the remaining k coordinates in $[-B,B]$, the maximum of*

$$\frac{|\mathbb{E}\chi_{[n]}(X)\tanh(u \cdot X)|}{\|u\|_2}$$

is exactly $q_n(a_n)/(a_n\sqrt{n})$. Its only maximizers have $a = a_n$ and every exceptional coordinate equal to a_n or $-a_n$. The threshold for n may depend on k and B.

Proof. Let $r = n - k$ and $\eta = r \bmod 2$. Central-binomial normalization of the equal bulk, followed by averaging the k exceptional signs, gives the leading phase response

$$H(a,b) = \mathbb{E}_Y \chi_{[k]}(Y) T_a\left(\eta a + \sum_i b_i Y_i\right).$$

There is a uniform expansion through order n^{-2}, with every fixed number of derivatives, on the stated compact parameter set. Here are explicit ways to justify it. For even r, pair the bulk levels $\pm 2j$. The normalized expression is

$$\tanh c + \sum_{j\ge 1} (-1)^j R_j(r)[\tanh(c+2aj) + \tanh(c-2aj)], \qquad R_j(r) = \frac{\binom{r}{r/2+j}}{\binom{r}{r/2}}.$$

Expand $R_j(r) = 1 - 2j^2/r + O_j(r^{-2})$ and sum the polynomial remainders against the exponentially decreasing paired bracket. For odd r, pair the levels $\pm(2j+1)$ and subtract the limiting bracket two. The remainder and all fixed derivatives again decay exponentially. The constant part sums exactly to the majority central-binomial coefficient. Splitting j at a small fixed multiple of r, as in the proof of Theorem 7, makes all remainders uniform.

The leading objective, after multiplication by n, is bounded by $\sqrt{2/\pi}\, F_0(a)/a$. Theorem 21 says that equality requires every b_i/a to be an odd integer. The limiting gain has the unique maximizer $a_0 \in (1,2)$ from the balanced expansion. Its second derivative is strictly negative; this also follows by passing the strict alternating-term estimate in the balanced-gain proof to the limit. There are only finitely many equality patterns in the compact box.

Near such a pattern, write $b_i = r_i a + e_i$, with odd integers r_i. Up to one common phase, the leading response is $\mathbb{E}T_a(a + \sum_i e_i Y_i)$. Its transverse Hessian at zero is $T_a''(a) I$, because $\mathbb{E}Y_i Y_j = 0$ for $i \ne j$. Its gain–transverse cross derivatives vanish. Thus every leading maximum is nondegenerate. Uniform C^2 convergence localizes all finite-n maximizers to these finitely many neighborhoods; strict negative Hessians and the gradient expansion give one local maximum in each, displaced by $O(n^{-1})$. If a pattern lies on a boundary of the prescribed box, enlarge that local box slightly and use its unconstrained maximum as an upper bound.

Evaluating a harmonic pattern at gain a_n and using (48) gives the relative value $1 - \Delta/n + O(n^{-2})$. Moving to the nearby local maximum improves this by only $O(n^{-2})$, because its gradient is $O(n^{-1})$ and its Hessian is uniformly negative in the objective multiplied by n. Every $\Delta > 0$ therefore loses. For $\Delta = 0$, the balanced vector is exactly stationary and local uniqueness identifies the maximum. □

The fixed-dimensional theorem also gives the sharp upper bound when a fixed number of bounded outside-coordinate weights and a bounded bias are present, with their squared weights included in the denominator. Apply the bias identity (8) to absorb the bias into one selected coordinate, use Theorem 24 on the two resulting vectors, and use concavity of the square root. Conditioning the finitely many outside signs and applying Jensen once more includes their squared norm.

The estimates in this appendix are inputs to the global argument. Appendix G combines them with lattice localization and residual asymptotics to handle arbitrarily many exceptional coordinates. The code theorem then adds orientation localization and high-order absorption.

D A global odd-lattice localization theorem

The following result applies to arbitrary incoming magnitudes. It is a necessary condition for competitive parity correlation, not the sharp atomic inequality.

Theorem 25 (Constant residual energy near an odd lattice)**.** *For every $c > 0$ there are constants $0 < a_- < a_+ < \infty$, $K < \infty$, and n_c such that the following holds for odd $n \geq n_c$. If $u \in \mathbb{R}^n \setminus \{0\}$ satisfies*

$$\frac{\left|\mathbb{E}\chi_{[n]}(X)\tanh\left(\sum_i u_i X_i\right)\right|}{\|u\|_2} \geq \frac{c}{n},$$

then there is $a \in [a_-, a_+]$ with

$$\sum_{i=1}^{n} \operatorname{dist}(|u_i|, (2\mathbb{Z}+1)a)^2 \leq K.$$

Moreover $\|u\|_2$ lies between two positive constant multiples of $\sqrt{n}$. If r_i are nearest positive odd integers, so that $||u_i| - ar_i| = \operatorname{dist}(|u_i|, (2\mathbb{Z}+1)a)$, then $\sum_i r_i^2 = O_c(n)$. All constants are independent of n.

Proof. Input sign changes permit assuming $u_i \geq 0$. A zero weight makes the parity coefficient zero, so under the hypothesis all weights are positive. Write that coefficient as $C_n(u)$ and put

$$r_p(a) = \int_0^\infty \frac{|\sin(at)|^p}{\sinh(\pi t/2)}\,dt, \qquad B_p = \int_0^\pi \sin(\theta)^p\,d\theta \qquad (p \geq 3).$$

We first prove the uniform comparison

$$r_p(a) \leq \frac{B_p}{B_3} r_3(a) \qquad (a > 0,\ p \geq 3). \tag{49}$$

After the substitution $\theta = at$ and folding consecutive periods, the kernel on $(0, \pi)$ is

$$K_a(\theta) = \sum_{j \geq 0} \frac{1}{\sinh(\pi(\theta + j\pi)/(2a))}.$$

It is decreasing and convex. On $(0, \pi/2)$ its symmetrization $L_a(\theta) = K_a(\theta) + K_a(\pi - \theta)$ is decreasing, because $L_a'(\theta) = K_a'(\theta) - K_a'(\pi - \theta) \leq 0$. Under the probability measure proportional to $\sin^3\theta\,d\theta$ on this interval, $\sin^{p-3}\theta$ is increasing and L_a is decreasing. Their covariance is therefore nonpositive. Dividing the resulting integral inequality by the positive integral of L_a gives (49). The inverse-linear endpoint singularity is integrable against $\sin^3\theta$, so truncation justifies the covariance argument directly.

The beta integral implies $\sup_{p \geq 3} \sqrt{p}\, B_p < \infty$. Hence, for an absolute constant A_0,

$$g_n(a) := \frac{\sqrt{n}\, r_n(a)}{a} \leq H(a) := A_0 \frac{r_3(a)}{a}.$$

The continuous function H is bounded and tends to zero as $a \downarrow 0$ and as $a \to \infty$. Indeed $|\sin(at)|^3 \leq a^3 t^3$ gives $r_3(a)/a = O(a^2)$ at zero. At infinity, splitting the integral at $1/a$ and 1, using $1/\sinh(\pi t/2) \leq 2/(\pi t)$ near zero and exponential decay beyond one, gives $r_3(a) = O(1 + \log a)$.

The exact sine representation of tanh, Hölder's inequality, and the geometric-mean–root-mean-square inequality give

$$|C_n(u)| \leq \prod_i r_n(u_i)^{1/n}, \qquad \frac{n|C_n(u)|}{\|u\|_2} \leq \prod_i g_n(u_i)^{1/n}.$$

Choose a bound $G > c$ for H and set $\delta = c^2/(2G)$. If fewer than $n/2$ of the values $g_n(u_i)$ were at least δ, their geometric mean would be at most $\sqrt{G\delta} = c/\sqrt{2}$, a contradiction. Thus at least $n/2$ weights lie in a fixed compact interval $[b, B] \subset (0, \infty)$ containing $\{a : H(a) \geq \delta\}$.

A coordinatewise nondecreasing function from the cube to $[-1, 1]$ has absolute top Fourier coefficient at most

$$\kappa_n = 2^{1-n} \binom{n-1}{(n-1)/2} = O(n^{-1/2}).$$

For completeness, average over coordinate permutations and express the resulting increasing sequence on Hamming levels as a convex combination of level thresholds and a constant. The alternating binomial identity bounds each threshold coefficient by the displayed central value. Applying this to the tanh ridge and using its assumed gain gives $\|u\|_2 \leq A\sqrt{n}$ for a constant $A = A(c)$. The moderate-coordinate count gives $\|u\|_2 \geq b\sqrt{n/2}$, and therefore

$$|C_n(u)| \geq \frac{\beta}{\sqrt{n}}, \qquad \beta = cb/\sqrt{2} > 0. \tag{50}$$

Put $P(t) = \prod_i |\sin(u_i t)|$ and $k(t) = 1/\sinh(\pi t/2)$. Let J index the moderate coordinates and write $N = |J| \geq n/2$. On every interval I of length one, uniformly in its location, for $v \in [b, B]$ and $p \geq 1$,

$$\int_I |\sin(vt)|^p \, dt \leq \frac{C(b, B)}{\sqrt{p}}.$$

This follows by substitution and covering the resulting interval by a bounded number of full or partial periods, using $B_p = O(p^{-1/2})$. Hölder over the coordinates in J consequently gives

$$\int_I P(t) \, dt \leq \frac{C_1}{\sqrt{n}}, \qquad \int_I P(t)^{1/2} \, dt \leq \frac{C_2}{\sqrt{n}}. \tag{51}$$

The second bound uses exponent $N/2$ in the single-coordinate integrals. For sufficiently large n this exponent is at least one.

Summing unit intervals and using exponential decay of k shows that

$$\int_T^\infty P(t)k(t) \, dt \leq \frac{C_3 e^{-\pi T/2}}{\sqrt{n}} \qquad (T \geq 1).$$

Choose a fixed T making this at most $\beta/(4\sqrt{n})$. Near zero, arithmetic–geometric mean gives

$$P(t) \leq t^n \prod_i u_i \leq (At)^n.$$

Choose a fixed $0 < t_0 < \min\{1, 1/(2A)\}$. Since $k(t) \leq 2/(\pi t)$, the integral on $(0, t_0)$ is exponentially small in n, and is at most $\beta/(4\sqrt{n})$ for large n. The sine representation and (50) now imply

$$\int_{t_0}^T P(t)k(t) \, dt \geq \frac{\beta}{2\sqrt{n}}.$$

Let $M = \max_{[t_0, T]} P$. Pointwise $P \leq \sqrt{M} \, P^{1/2}$. Because k is bounded on this compact interval, the second estimate in (51) gives

$$\frac{\beta}{2\sqrt{n}} \leq \int_{t_0}^T Pk \leq \frac{C_4\sqrt{M}}{\sqrt{n}}.$$

Thus $M \ge \mu := (\beta/(2C_4))^2 > 0$, independently of n. Choose $t_* \in [t_0, T]$ with $P(t_*) \ge \mu$, and let $\rho_i = \operatorname{dist}(u_i t_*, \pi/2 + \pi\mathbb{Z}) \in [0, \pi/2]$. Since $-\log\cos\rho \ge \rho^2/2$,

$$\sum_i \rho_i^2 \le -2\log P(t_*) \le -2\log\mu.$$

Taking $a = \pi/(2t_*) \in [\pi/(2T), \pi/(2t_0)]$ and dividing by t_*^2 proves the asserted constant residual bound. Finally,

$$a^2 \sum_i r_i^2 \le 2\|u\|_2^2 + 2K = O_c(n),$$

which gives the remaining conclusion. □

The theorem does not force the residual energy to vanish, nor does it bound the number of nontrivial odd harmonics. A sharp global atom theorem requires a uniform signed comparison inside this localized family.

E Uniform signed asymptotics for a lattice bulk

The next estimate is uniform in an arbitrary external affine phase. This uniformity permits conditioning on coordinates outside a large bulk, including coordinates with unbounded individual weights. For $E \ge 0$ define the Gaussian smoothing

$$T_{A,E}(b) = \mathbb{E}_Z T_A(b + \sqrt{E}Z), \qquad Z \sim N(0,1).$$

Here T_A is the periodic response in (42).

Theorem 26 (A finite lattice alphabet with residual energy)**.** *Fix a finite set of positive odd integers with greatest common divisor one, a compact interval $\mathcal{I} \subset (0,\infty)$, and constants $K < \infty$ and $\alpha > 0$. For each N, let $q_1, \dots, q_N$ belong to this alphabet, with each alphabet member occurring at least αN times. Let $A \in \mathcal{I}$ and*

$$u_i = Aq_i + d_i, \qquad V = \sum_i q_i^2, \qquad E = \sum_i d_i^2 \le K, \qquad \sum_i q_i d_i = 0.$$

Suppose $\max_i |d_i| \to 0$ as $N \to \infty$, and put $\eta = N \bmod 2$. There is a sign $\epsilon_N \in \{-1, 1\}$ such that, uniformly for all $B \in \mathbb{R}$,

$$\mathbb{E}_X \chi_{[N]}(X) \tanh\left(B + \sum_i u_i X_i\right) = \epsilon_N \sqrt{\frac{2}{\pi V}} e^{-B^2/(2A^2V)} T_{A,E}(B + \eta A) + o(V^{-1/2}). \tag{52}$$

The assertion remains valid after any fixed number of derivatives in B, with the leading expression differentiated. The remainders are uniform over the displayed compact gain interval, residual-energy bound, and lower bound on alphabet frequencies.

Proof. The exact sine representation gives

$$\mathbb{E}\chi_{[N]} \tanh\left(B + \sum_i u_i X_i\right) = \int_0^\infty \frac{\sin(Bt + N\pi/2) \prod_i \sin(u_i t)}{\sinh(\pi t/2)}\, dt. \tag{53}$$

The ideal product with $d_i = 0$ has simultaneous absolute maxima exactly at

$$t_l = \frac{(2l+1)\pi}{2A}, \qquad l = 0, 1, \dots.$$

Indeed all q_i are odd; conversely, if $q_i tA$ is an odd multiple of $\pi/2$ for every alphabet member, their greatest common divisor one forces $2At/\pi$ to be an odd integer.

Fix a bounded t interval. Near t_l, write $t = t_l + x$ and factor out the exact sine signs. The remaining product is

$$\prod_i \cos z_i, \qquad z_i = Aq_i x + d_i(t_l + x).$$

The orthogonality assumption gives the exact identity

$$\sum_i z_i^2 = A^2 V x^2 + E(t_l + x)^2.$$

Choose $L_N \to \infty$ slowly enough that $L_N^4/N \to 0$ and $(\max_i |d_i|)^2 L_N^2 \to 0$. For example, take the minimum of $N^{1/8}$ and $(\max_i |d_i|)^{-1/2}$, omitting the second restriction when all residuals vanish. On $|x| \le L_N/\sqrt{N}$,

$$\max_i |z_i| \to 0, \qquad \sum_i z_i^4 \le \max_i z_i^2 \sum_i z_i^2 = o(1).$$

The Taylor expansion $\log\cos z = -z^2/2 + O(z^4)$ consequently gives, uniformly on these windows,

$$\prod_i \sin(u_i t) = \epsilon'_N (-1)^{Nl} e^{-A^2 V x^2/2} e^{-Et_l^2/2}(1 + o(1)). \tag{54}$$

The cross terms $2Et_l x + Ex^2$ are $o(1)$ on the windows. The error integrated against the Gaussian is $o(N^{-1/2})$.

We justify discarding the rest of the fixed compact interval. The inequality $(x+y)^2 \ge x^2/2 - y^2$, together with the Lipschitz bound for cosine, gives

$$\sum_i \cos^2(u_i t) \ge \frac{1}{2}\sum_i \cos^2(Aq_i t) - t^2 E.$$

The finite alphabet and its positive frequency lower bounds imply that the ideal sum on the right is at least cN times the squared distance to the common peak set near those peaks, and at least cN away from fixed peak neighborhoods. This follows by compactness and the nonzero quadratic term at each common zero of the cosines. Since

$$\prod_i |\sin(u_i t)| \le \exp\left(-\frac{1}{2}\sum_i \cos^2(u_i t)\right),$$

the integral outside the growing Gaussian windows is $o(N^{-1/2})$. Near zero, all amplitudes remain in a common compact positive interval, so the product vanishes to order N and the inverse-linear kernel singularity causes no difficulty.

Freeze the smooth kernel $1/\sinh(\pi t/2)$ at each peak. The resulting L^1 error on the fixed compact interval is $o(N^{-1/2})$. Multiplication by e^{iBt} leaves its magnitude unchanged, so this error estimate is uniform in every real B. The exact Gaussian Fourier transform contributes

$$\int_{\mathbb{R}} e^{-A^2 V x^2/2} e^{iBx}\, dx = \frac{\sqrt{2\pi}}{A\sqrt{V}} e^{-B^2/(2A^2 V)}.$$

The factor $e^{-Et_l^2/2}$ in (54) is precisely Gaussian smoothing of variance E in the external phase.

For clarity, the Fourier series of the periodic response is

$$T_A(b) = \frac{\pi}{A}\sum_{l\ge 0} \frac{\sin((2l+1)\pi b/(2A))}{\sinh((2l+1)\pi^2/(4A))}.$$

It follows either by Poisson summation of the absolutely convergent periodization of sech^2, or by differentiating the sine representation of tanh and integrating the resulting Fourier series. The factor $(-1)^{Nl}$ in (54) leaves the sine phases unchanged when N is even, and gives alternating cosine phases when N is odd. These are exactly the phases of $T_A(B+\eta A)$, up to one common sign. Combining the Gaussian constant with the factor π/A in this series gives $\sqrt{2/(\pi V)}$, as in (52).

It remains to control the infinite tail. All actual bulk amplitudes lie in a common compact positive interval for large N. On each unit t interval, Hölder and $\int_0^\pi |\sin\theta|^N \, d\theta = O(N^{-1/2})$ bound the product integral by $O(N^{-1/2})$, uniformly in the interval's location. The weighted tail beyond T is therefore $O(e^{-\pi T/2}/\sqrt{N})$. The model peak series has the same exponential tail. First taking $N \to \infty$ with T fixed and then $T \to \infty$ proves the uniform remainder. Multiplication by any fixed power of t proves the derivative statement: the tail gains only a polynomial factor in T. Equivalently, the Gaussian model can be Fourier transformed before differentiating; its differentiated expression has the same $o(V^{-1/2})$ error. □

The orthogonality condition costs no generality in applications. Replacing A by $A + \sum_i q_i d_i / V$ is least-squares refitting. It changes A by $O(N^{-1/2})$, decreases the residual energy, and preserves $\max_i |d_i| \to 0$ for a fixed alphabet.

Corollary 27 (Uniform external-phase bound)**.** *Under the hypotheses of Theorem 26,*

$$\sup_{B\in\mathbb{R}} \left| \mathbb{E}\chi_{[N]} \tanh\left(B + \sum_i u_i X_i \right) \right| \le \sqrt{\frac{2}{\pi V}} F_0(A) + o(V^{-1/2}).$$

If the residual energy is bounded below by a fixed positive number, Gaussian smoothing gives a uniform strict loss in the periodic factor, when A and E range over fixed compact intervals.

Proof. Theorem 20 gives $|T_A| \le F_0(A)$, and Gaussian averaging preserves this bound. If $E > 0$, the Gaussian has full support and averages a nonconstant periodic function with isolated extrema, so the inequality for its supremum is strict. Continuity and compactness make that loss uniform when E stays bounded away from zero. □

F A uniform bias bound for an equal bulk

The global atomic argument requires control of all possible biases after conditioning on exceptional coordinates. The next theorem supplies this control, including the second-order error needed to rule out a single exceptional coordinate.

For $r \ge 1$, define

$$B_r(a,b) = \mathbb{E}_X \left[\chi_{[r]}(X) \tanh\left(a \sum_{i=1}^{r} X_i + b \right) \right].$$

The normalization $q_n(a)$ refers to the positive balanced coefficient of odd degree n.

Theorem 28 (An arbitrary bias cannot improve an equal bulk)**.** *Uniformly for $a \in [1,2]$, the following hold for all sufficiently large r.*

1. *If r is odd, then*

$$\sup_{b\in\mathbb{R}} |B_r(a,b)| = q_r(a),$$

and the unique maximizing bias is $b = 0$.

2. If r is even, then

$$\sup_{b\in\mathbb{R}} |B_r(a,b)| \le q_{r+1}(a)\,[1+O(r^{-2})].$$

Every maximizing bias lies at $b = a + O(r^{-1})$ or $b = -a + O(r^{-1})$.

All constants and the sufficiently-large threshold are independent of a.

Proof. We first prove that every maximizing bias lies in a fixed compact interval. This step is needed before using a local expansion around the periodic maxima.

Put

$$K_a(x) = \frac{\tanh(x+a) - \tanh(x-a)}{2} > 0.$$

The kernel and every fixed derivative decrease exponentially in $|x|$, uniformly for $a \in [1,2]$. Taking one selected finite difference first gives the exact identity

$$B_r(a,b) = \mathbb{E}_{X\in\{-1,1\}^{r-1}} \chi_{[r-1]}(X) K_a\left(a\sum_i X_i + b\right). \tag{55}$$

Its alternating periodization is

$$\sum_{j\in\mathbb{Z}} (-1)^j K_a(c+2aj) = T_a(c+a). \tag{56}$$

Indeed K_a is the difference of two shifted tanh functions, and $T_a(c-a) = -T_a(c+a)$.

Write $N = r-1$. Let p_j be the probabilities of $\frac{1}{2}\sum_{i=1}^N X_i$, on the integer lattice when N is even and on the half-integer lattice when N is odd. Put $p_* = \max_j p_j$. Up to a common sign, (55) is

$$\sum_j (-1)^{j-j_0} p_j K_a(b+2aj),$$

where j_0 is any fixed element of the mass lattice. Choose an index l of that lattice nearest $-b/(2a)$ and put $c = b+2al \in [-a,a]$. After shifting the summation index, the coefficient is, up to a common sign,

$$\sum_{j\in\mathbb{Z}} (-1)^j p_{l+j} K_a(c+2aj). \tag{57}$$

We record the binomial estimates used below. For a sufficiently small absolute $\varepsilon > 0$ and $|l| \le \varepsilon N$,

$$p_l/p_* \le \exp\left[-c_1(|l|-1)_+^2/N\right], \tag{58}$$

$$\sum_j \left|\frac{p_{l+j}}{p_l} - 1\right| |K_a(c+2aj)| \le C\frac{|l|+1}{N}. \tag{59}$$

The constants apply to both lattice parities. To check the second estimate directly, use the consecutive ratio

$$\frac{p_{s+1}}{p_s} = \frac{N/2-s}{N/2+s+1}.$$

Summing logarithms for $|l|, |j| \le \varepsilon N$ gives

$$\left|\frac{p_{l+j}}{p_l} - 1\right| \le C\frac{|l||j|+j^2+|j|}{N} \exp\left(C\frac{|l||j|+j^2+|j|}{N}\right).$$

Choose ε so that the last exponential is absorbed by the fixed exponential decay of K_a. For the remaining indices, $p_l \geq p_* e^{-C\varepsilon^2 N}$ and the kernel tail is $O(e^{-c\varepsilon N})$; decreasing ε absorbs their ratio and proves (59). The first estimate follows from the same consecutive ratios, using $\log(1-x) \leq -x$.

The periodic maximum $|T_a| \leq F_0(a)$, together with (57)–(59), now gives

$$\frac{|B_r(a,b)|}{p_*} \leq e^{-c_1(|l|-1)_+^2/N} \left[F_0(a) + C\frac{|l|+1}{N}\right] \qquad (|l| \leq \varepsilon N). \tag{60}$$

For $|l| > \varepsilon N$, split (57) at $|j| = |l|/2$. The first part uses the Gaussian binomial bound $p_{l+j} \leq p_* e^{-cl^2/N}$, and the second uses the exponentially small kernel tail. The result is exponentially small in N; the same argument applies if l is outside the binomial support.

There is a central candidate of size $p_*[F_0(a) - O(N^{-1})]$: use $b = 0$ for odd r and $b = a$ for even r. Since F_0 is uniformly positive on $[1,2]$, (60) confines every maximizing index to $|l| \leq L$ for an absolute constant L. More explicitly, for $L \leq |l| \leq \sqrt{N}$, the negative quadratic term $-cl^2/N$ dominates the linear correction $C(|l|+1)/N$ once L is large. For $\sqrt{N} \leq |l| \leq \varepsilon N$, reduce ε so that the fixed Gaussian loss dominates that correction. All remaining indices were already exponentially small. Thus every maximizing bias lies in a fixed compact interval, uniformly in a and r.

On that compact interval, expand the binomial probabilities in (55) around the center. The localized kernel allows the polynomial-in-j remainders to be summed with any fixed number of bias derivatives. This gives a uniform expansion through $O(r^{-2})$, after central-binomial normalization. Its leading response is $T_a(b+a)$ for odd r and $T_a(b)$ for even r, up to a common sign. Their maxima are nondegenerate. Consequently every maximizing bias lies within $O(r^{-1})$ of one of finitely many periodic peaks, and optimizing within such a neighborhood changes its normalized value by only $O(r^{-2})$.

For odd r, the peaks are $b = 2la$. The Fourier identity for $\cos(2l\theta)$ and the balanced ratio

$$\frac{q_{r+2j}(a)}{q_r(a)} = 1 - \frac{j}{r} + O(r^{-2})$$

give, uniformly over the fixed range of l,

$$\frac{|B_r(a,2la)|}{q_r(a)} = 1 - \frac{2l^2}{r} + O(r^{-2}). \tag{61}$$

One way to see the coefficient is to normalize the polynomial $\cos(2l\theta)$ so that its value at $y = \sin^2\theta = 1$ is one; its derivative there is $2l^2$. Every nonzero l loses even after the $O(r^{-2})$ local improvement. At $l = 0$, the response is even in b, so zero is exactly stationary; strict local concavity makes it the unique local, hence global, maximum of the absolute coefficient.

For even r, set $n = r+1$. The peaks are $b = (2l+1)a$. The identity $B_r(a,b) = C_n(a,\ldots,a,b)$ identifies them with one odd harmonic in an odd-degree parity atom. The sine polynomial therefore gives

$$\frac{|B_r(a,(2l+1)a)|}{q_n(a)} = 1 - \frac{2l(l+1)}{n} + O(n^{-2}). \tag{62}$$

Only $l = 0, -1$, corresponding to $b = \pm a$, survive. Local optimization shifts them by $O(r^{-1})$ and increases their relative value by $O(r^{-2})$, as required. □

G The sharp global single-parity bound

Write

$$C_n(u) = \mathbb{E}\chi_{[n]}(X)\tanh(u\cdot X), \qquad L(a) = \sqrt{2/\pi}\,\frac{F_0(a)}{a}, \qquad L_* = L(a_0).$$

Here $a_0 \in (1,2)$ is the limiting balanced gain from Theorem 7. It is the unique global maximizer of L. Here the strictness survives the limit quantitatively. For $a \le 1$, the positive-series proof gives

$$a(q_n/a)' \ge 2\delta\, q_n/a, \qquad \delta = 44428/44251 - 1 > 0.$$

Divide by κ_n and pass to the locally uniform limit to obtain $a(F_0/a)' \ge 2\delta F_0/a > 0$. For $a \ge 1$, the alternating-series proof of concavity has a uniformly positive first-pair gap: with $p(t) = -\tanh'' t > 0$, its normalized limit obeys

$$F_0''(a) \le -2[p(a) - 9p(3a)] < 0.$$

The inequality $9p(3a) < p(a)$ is the explicit ratio bound in Theorem 4. Thus $aF_0' - F_0$ is strictly decreasing above one. Its positive sign at one and negative sign at two also follow from the strict normalized endpoint bounds in that theorem. There is exactly one zero $a_0 \in (1,2)$, and the limiting scalar maximum is nondegenerate. We first locate every asymptotically optimal unrestricted neuron.

Lemma 29 (Asymptotic localization at one gain)**.** *Along odd integers $n \to \infty$,*

$$n \sup_{u \ne 0} \frac{|C_n(u)|}{\|u\|_2} \longrightarrow L_*.$$

Every sequence with scaled gain at least $L_* - o(1)$ satisfies, after input sign changes,

$$\frac{1}{n}\#\{i : |u_i - a_0| > \epsilon\} \longrightarrow 0 \quad (\epsilon > 0), \qquad \frac{\|u\|_2^2}{n} \longrightarrow a_0^2.$$

After taking a subsequence there are $A_n \to a_0$ and sets I_n of size $n - o(n)$ such that

$$\max_{i \in I_n} |u_i - A_n| \to 0, \qquad \sum_{i \in I_n} (u_i - A_n)^2 \le K, \qquad \sum_{i \notin I_n} u_i^2 = o(n). \tag{63}$$

The constant K is absolute for any fixed positive competitive-gain threshold below L_*.

Proof. Balanced neurons give the lower bound. Consider any sequence with scaled gain bounded below by a positive constant. By Theorem 25, after input sign changes,

$$u_i = a_n r_i + d_i, \qquad r_i \in \{1,3,5,\dots\}, \qquad a_n \in [a_-, a_+], \qquad \sum_i d_i^2 \le K, \quad \sum_i r_i^2 \le K_1 n.$$

Pass to a subsequence with $a_n \to a$, with empirical probabilities of each odd integer converging to p_r, and with $n^{-1}\sum_i r_i^2 \to Q_{\text{total}}$. The second-moment bound makes the empirical laws tight, so $\sum_r p_r = 1$. Put

$$Q_{\lim} = \sum_r p_r r^2 \le Q_{\text{total}}.$$

The bounded residual energy implies $\|u\|_2^2/n \to a^2 Q_{\text{total}}$.

Fix a finite cutoff J containing a type of positive limiting frequency. Retain the coordinates of types $r \le J$ with $p_r > 0$, and discard $o(n)$ further coordinates so their maximum residual tends to zero. This is possible by deleting $|d_i| > \epsilon_n$, where $\epsilon_n \downarrow 0$ sufficiently slowly that $K/\epsilon_n^2 = o(n)$. Let g_J be the gcd of the retained types and $Q_J = \sum_{r \le J} p_r r^2$. Least-squares refitting their common scale changes $a_n g_J$ by $O(n^{-1/2})$, decreases residual energy, and preserves the vanishing maximum residual. Condition on every other selected coordinate. The uniform-in-bias residual central limit theorem, Theorem 26, and $\|T_{A,E}\|_\infty \le F_0(A)$ give

$$|C_n(u)| \le \sqrt{\frac{2}{\pi}}\, \frac{F_0(a g_J)}{\sqrt{n Q_J / g_J^2}} + o(n^{-1/2}).$$

The outside parity sign does not affect this upper bound. Therefore

$$\limsup_n \frac{n|C_n(u)|}{\|u\|_2} \le L(ag_J)\frac{g_J^2}{\sqrt{Q_J Q_{\text{total}}}}. \tag{64}$$

As J increases through the support of p, its positive integer gcds eventually stabilize at a value g. Also $Q_J \uparrow Q_{\text{lim}}$. Every support point is a positive odd multiple of g, whence $Q_{\text{total}} \ge Q_{\text{lim}} \ge g^2$. Equation (64) proves the upper bound L_*. Equality at this bound forces

$$ag = a_0, \qquad Q_{\text{total}} = Q_{\text{lim}} = g^2.$$

Thus $p_g = 1$ and no second-moment mass escapes to larger harmonics. Retain the coordinates of type g with $|d_i| \le \epsilon_n$. They give (63) with $A_n = a_n g$. Applying this argument to every subsequence proves the asserted empirical and norm conclusions for the original sequence. □

Lemma 30 (An exactly equal majority at a global maximizer)**.** *For any sequence of global maximizers of $|C_n(u)|/\|u\|_2$ with odd $n \to \infty$, after input sign changes there is an exactly equal bulk of size $n - o(n)$, of gain tending to a_0. All remaining coordinates have total squared norm $o(n)$. Moreover, one fixed neighborhood of the common bulk gain contains no exceptional coordinate for all sufficiently large n.*

Proof. For each fixed $n \ge 3$ the ratio tends to zero at infinity. It also tends to zero at the origin, since the top Fourier coefficient vanishes to order at least n there. Thus a nonzero global maximum is attained. None of its selected coordinates is zero. Apply Lemma 29 and use its set I_n.

Choose $i, j \in I_n$, and put $\mu = (u_i + u_j)/2$ and $h = (u_i - u_j)/2$. The remaining selected degree $n - 2$ is odd, so its response

$$G(b) = \mathbb{E}\chi_{[n]\setminus\{i,j\}}(X)\tanh\left(b + \sum_{l\ne i,j} u_l X_l\right)$$

is even. Conditioning the two removed signs gives the exact identity

$$C_n(u) = \tfrac{1}{2}\{G(2\mu) - G(2h)\}. \tag{65}$$

Remove this pair from I_n and least-squares refit its common gain a_n. Its residual energy stays bounded, its maximum residual tends to zero, and $a_n \to a_0$. Let $N = n - o(n)$ be the remaining bulk size, let E be its residual energy, and write W for the sum of the other selected coordinates with their independent signs. Then $\mathbb{E}W^2 = o(n)$. Theorem 26, including two bias derivatives, gives

$$G(b) = \epsilon_N\sqrt{\frac{2}{\pi N}}\mathbb{E}\left[\chi_{\text{out}} e^{-(b+W)^2/(2a_n^2 N)} T_{a_n,E}(b + W + \eta a_n)\right] + o(n^{-1/2}), \qquad \eta = N \bmod 2.$$

The statement and its first two derivatives are uniform in b on compact sets and uniform over the removed pair. The Gaussian envelope can be removed in expectation there, since $\mathbb{E}W^2 = o(n)$; its first and second derivative contributions also tend to zero after multiplication by $\sqrt{n}$. The periodic derivatives are uniformly bounded. Thus, for $j = 0, 1, 2$,

$$G^{(j)}(b) = \epsilon_N\sqrt{\frac{2}{\pi N}}\mathbb{E}[\chi_{\text{out}} T^{(j)}_{a_n,E}(b + W + \eta a_n)] + o(n^{-1/2}). \tag{66}$$

Every parity sign can be absorbed into a shift by $2a_n$, using anti-periodicity. Put $s = \operatorname{sign} C_n(u)$ and $H = -sG$. Consequently (66) represents H and its derivatives as probability averages of the same smoothed periodic response, with positive normalizing factor $\sqrt{2/(\pi N)}$.

Since $\mu - a_n \to 0$ and $h \to 0$, anti-periodicity gives $G(2\mu) = -G(0) + o(n^{-1/2})$ and $G(2h) = G(0) + o(n^{-1/2})$. Global optimality and (65) therefore force the normalized average representing $H(0)$ to tend to $F_0(a_0)$. Choose a fixed small $\delta > 0$. Uniform boundedness of the first periodic derivative then makes this average at least $4F_0(a_n)/5$ throughout $|b| \le \delta$, for all large n and uniformly in the pair.

Lemma 22 is preserved by Gaussian smoothing and probability averaging. It and (66) therefore give one $c > 0$ with

$$H''(b) \le -c/\sqrt{n} \qquad (|b| \le \delta).$$

Keep the pair mean fixed and vary its half-difference x toward zero. Its positive signed coefficient is $q(x) = \frac{1}{2}\{H(2x) - H(2\mu)\}$. Evenness gives $q'(0) = 0$, and $q''(x) = 2H''(2x) < 0$ throughout the relevant segment. Unless $h = 0$, balancing the pair strictly increases the coefficient while strictly decreasing its squared norm $\|u_{\text{rest}}\|_2^2 + 2\mu^2 + 2x^2$. This contradicts global maximality. All coordinates of I_n are therefore exactly equal.

The same proof includes one additional coordinate within a fixed sufficiently small distance δ_0 of the common gain. Its pair mean and half-difference change the normalized response by $O(\delta_0)$; removing it from the outside sum preserves $\mathbb{E}W^2 = o(n)$. Choose δ_0 so that the $4/5$ peak threshold and the curvature interval still apply. Such a coordinate must equal the bulk gain too. This proves the final assertion. □

Theorem 31 (Sharp global affine tanh parity bound)**.** *For every sufficiently large odd n, and every affine tanh atom on any finite Boolean cube containing a selected set A of size n,*

$$w_n \, |\mathbb{E}\chi_A(X) \tanh(b + u \cdot X)| \le \sqrt{b^2 + \|u\|_2^2}.$$

At nonzero equality, the bias and all coordinates outside A vanish, and the selected coordinate magnitudes all equal the unique optimal balanced gain a_n.

Proof. First consider the bias-free intrinsic problem. Suppose nonbalanced global maximizers exist for arbitrarily large odd n. By Lemma 30, write their exactly equal bulk as $N = n - k$ coordinates of gain $a \to a_0$, with $k = o(n)$ and outside weights w_i satisfying $\sum_i w_i^2 = o(n)$. Every exceptional coordinate is outside one fixed neighborhood of a.

Conditioning on the k exceptional signs and applying the equal-bulk case of Theorem 26 gives, with $B = \sum_i w_i Y_i$ and $\eta = N \bmod 2$,

$$C_n = \epsilon_N \sqrt{\frac{2}{\pi N}} \, \mathbb{E}\left[\chi_{[k]}(Y) e^{-B^2/(2a^2N)} T_a(B + \eta a)\right] + o(n^{-1/2}).$$

Since $\mathbb{E}B^2 = o(n)$ the Gaussian envelope can be removed. Global competitiveness and norm asymptotics imply

$$\left|\mathbb{E}\chi_{[k]}(Y) T_a\left(\eta a + \sum_i w_i Y_i\right)\right| \longrightarrow F_0(a).$$

The integers η and k have odd sum. The dimension-free phase stability theorem, Theorem 21, yields

$$\sum_i \operatorname{dist}(w_i, (2\mathbb{Z}+1)a)^2 \longrightarrow 0.$$

The fixed-neighborhood conclusion of Lemma 30 excludes the first positive odd multiple for every outlier. Hence $w_i \ge 3a - o(1)$ uniformly over them, and

$$\|u\|_2^2 \ge a^2\{n + (8 - o(1))k\}. \tag{67}$$

Put $\delta = 0$ if N is odd and $\delta = 1$ otherwise. Theorem 28 gives

$$|C_n| \le q_{N+\delta}(a)(1 + O(N^{-2})),$$

where the factor is absent for odd N. The balanced scalar expansion, uniform on $a \in [1, 2]$, gives for $h = o(n)$ with $n - h$ odd

$$\frac{q_{n-h}(a)}{q_n(a)} = \sqrt{\frac{n}{n-h}}\{1 + O(h/n^2)\}.$$

For $h = 0$ the ratio is exactly one. Taking $h = k - \delta$ and using $k \ge 1$, the even-bulk error is absorbed into $O(k/n^2)$. Thus

$$\frac{|C_n|/\|u\|_2}{q_n(a)/(a\sqrt{n})} \le \frac{n}{\sqrt{(n-k+\delta)(n+(8-o(1))k)}}\{1 + O(k/n^2)\} < 1$$

for all sufficiently large n. This contradicts global maximality, since the denominator is the gain of a balanced competitor. Therefore every global intrinsic maximizer is balanced, and Theorem 4 identifies its gain as a_n.

For a fixed bias B and any selected coordinate i, the odd-degree identity (8) reads

$$C_n(B; u) = \tfrac{1}{2}\{C_n(0; u + Be_i) + C_n(0; u - Be_i)\}.$$

The intrinsic bound and square-root concavity imply $w_n|C_n(B; u)| \le \sqrt{\|u\|_2^2 + B^2}$. Conditioning outside coordinates and applying Jensen proves the stated affine bound. At a nonzero equality every selected coordinate is nonzero. The square-root concavity step is strict whenever $B \ne 0$, because the two squared norms differ by $4u_iB$. Conditioning therefore forces $b + u_{A^c} \cdot X = 0$ at every outside input, hence zero bias and outside weights. Intrinsic equality gives the asserted selected gains. □

The support estimate needed for the code theorem is Lemma 8. It follows directly from this global intrinsic inequality and the quantitative square-root concavity gap; no quantitative stability of the selected balanced weights is needed.

H Localization to one transversal orientation

This appendix treats arbitrary coordinate weights, without assuming that they are constant on a rail. There is one shared carrier coordinate and $m \ge 2$ disjoint pairs of even rails. Every rail size belongs to $[R, R + \Delta]$, where $R \ge 4$ is even and $0 \le \Delta \le R$. A full transversal support consists of the carrier and one rail from every pair. Any retained subfamily $\mathcal{S}$ is allowed. Put $D = mR$ and

$$Q(h_u) = \sum_{s \in \mathcal{S}} w_{n_s} |\widehat{h}_u(A_s)|, \qquad h_u(x) = \tanh(u \cdot \bar{x}), \qquad n_s = |A_s|.$$

The incoming norm includes the bias and any irrelevant coordinates. The balanced scalar estimates give $w_n \le Cn$ for odd $n \ge 3$; hence $w_{\max} \le CD$ here.

Theorem 32 (A single retained orientation carries the mass)**.** *Fix $\beta > 0$. There are positive constants c, C, K_0, depending only on β, with the following property. Suppose $u \ne 0$, $R \ge K_0 \log^2(m+2)$, and $Q(h_u) \ge \beta\|u\|_2$. Then*

$$c\sqrt{D} \le \|u\|_2 \le C\sqrt{D},$$

and there is a retained orientation s_* such that

$$\sum_{s\neq s_*} |\widehat{h}_u(A_s)| \le Cme^{-cR} + \frac{C(1+\sqrt{R})e^{-c\sqrt{R}}}{\sqrt{D}}. \tag{68}$$

The sum may run over the whole full code, so the same bound holds on the retained subfamily. For every fixed $L > 0$, there is a constant $K_L \ge K_0$ such that all sufficiently large m satisfy, uniformly in $R \ge K_L \log^2(m+2)$,

$$\sum_{\substack{s\in\mathcal{S}\\ s\neq s_*}} w_{n_s}|\widehat{h}_u(A_s)| \le D^{-L}\|u\|_2. \tag{69}$$

Both K_L and the threshold for m may depend on L and β. Only the fixed choice $L = 8$ is needed in the exact code-norm theorem.

Proof. Input sign changes allow assuming nonnegative coordinate magnitudes. For pair i define

$$A_i(t) = \prod_{j\in B_i^0} |\sin(u_j t)| \prod_{j\in B_i^1} |\cos(u_j t)|, \qquad B_i(t) = \prod_{j\in B_i^0} |\cos(u_j t)| \prod_{j\in B_i^1} |\sin(u_j t)|,$$

and put $F(t) = \prod_i (A_i(t) + B_i(t))$. Write $F_{\mathcal{S}}$ for the corresponding sum of products over retained orientations. The coefficient integral for an odd support contains a common carrier sine, a bias cosine, and irrelevant-coordinate cosines. Taking absolute values therefore gives

$$\sum_{s\in\mathcal{S}} |\widehat{h}_u(A_s)| \le \int_0^\infty \frac{|\sin(u_0 t)| F_{\mathcal{S}}(t)}{\sinh(\pi t/2)}\, dt. \tag{70}$$

Here u_0 is the carrier magnitude. Each $A_i + B_i \le 1$: keep one coordinate from each rail, discard the other factors, and apply $|\sin x \cos y| + |\cos x \sin y| \le 1$. Thus $0 \le F \le 1$.

Uniform scale bounds. Choose any R coordinates from each rail and let L_0 be their combined Euclidean norm. Put $\lambda = L_0/\sqrt{D}$. We claim

$$\int_0^\infty \frac{F(t)}{\sinh(\pi t/2)}\, dt \le \frac{C}{\sqrt{D}} \min\{\lambda^3, \log(1+\lambda)\}. \tag{71}$$

Discard rail factors beyond the chosen coordinates and zip each pair into R columns. Generalized Hölder over the orientation index gives an upper bound by the geometric mean of the column functions

$$F_k(t) = \prod_{i=1}^m \left[\sin(a_i t)^R \cos(b_i t)^R + \cos(a_i t)^R \sin(b_i t)^R\right].$$

The notation a_i, b_i refers to a fixed column; the two magnitudes may be ordered separately in each column. Clarkson's even-power inequality bounds this function by

$$\mathbb{E}_\varepsilon \prod_i |\sin(c_i t)|^R, \qquad c_i = |a_i + \varepsilon_i b_i|.$$

Hölder over the m pairs bounds its integral by $\mathbb{E}_\varepsilon \prod_i r_D(c_i)^{1/m}$, where r_D is the absolute diagonal integral used in Appendix D. Equation (49) and the elementary bounds on r_3 give

$$r_D(c) \le \frac{C}{\sqrt{D}} r_3(c) \le \frac{C}{\sqrt{D}} \min\{c^3, \log(1+c)\}.$$

For the logarithmic bound, geometric mean at most arithmetic mean and concavity of $s \mapsto \log(1+\sqrt{s})$ imply

$$\mathbb{E}_\varepsilon \prod_i \log(1+c_i)^{1/m} \le \log\left(1+\sqrt{\frac{1}{m}\sum_i (a_i^2+b_i^2)}\right).$$

For the cubic bound use geometric mean at most root mean square and $\sum_i c_i^2 \le 2\sum_i (a_i^2+b_i^2)$ pointwise. Finally apply Hölder over the R columns. The geometric mean of the cubic bounds is at most the cube of the combined root mean square; for the logarithmic bounds use the same concavity once more. This proves (71).

Set $z = \|u\|_2/\sqrt{D}$. Since $L_0 \le \|u\|_2$, the functions in the right side of (71) are increasing, and $w_{\max} \le CD$. The competitiveness assumption yields

$$\beta z \le C \min\{z^3, \log(1+z)\}.$$

Consequently z belongs to a fixed compact interval $[z_-, z_+] \subset (0,\infty)$, proving the asserted norm bounds.

A constant-height resonance. For every $T \ge 0$, every frequency $v \ge 0$, and every $p \ge 1$,

$$\int_T^{T+1} |\sin(vt)|^p \, dt \le \frac{C(T+1)}{\sqrt{p}}. \tag{72}$$

If $v \le \pi/[4(T+1)]$, the sine is at most $1/\sqrt{2}$ there. Otherwise substitution and covering by full or partial periods proves the estimate, using $1/v \le 4(T+1)/\pi$. The column argument above also gives

$$\int_T^{T+1} F(t)^{1/2} \, dt \le \frac{C(T+1)}{\sqrt{D}}. \tag{73}$$

Indeed Hölder across columns reduces to a single column. Hölder across the m pairs then produces powers $m/2$ of a single pair factor. Since $m/2 \ge 1$, Clarkson followed by Jensen bounds these by averages of $|\sin((a_i \pm b_i)t)|^{D/2}$, to which (72) applies. As $F \le 1$, the same bound holds for F itself.

It follows that the weighted integral beyond $T \ge 1$ is at most $C(T+1)e^{-\pi T/2}/\sqrt{D}$. Near zero, every transversal uses at least D rail coordinates, and arithmetic–geometric mean gives

$$F(t) \le 2^m (z_+ t)^D \qquad (z_+ t \le 1).$$

A fixed small lower cutoff therefore makes the near-zero contribution exponentially small in D. On the other hand, (70), competitiveness, and $w_{\max} \le CD$ show that the retained positive integral is at least $c/\sqrt{D}$. Choose fixed $0 < t_- < t_+ < \infty$ so that

$$\int_{t_-}^{t_+} \frac{|\sin(u_0 t)| F_{\mathcal{S}}(t)}{\sinh(\pi t/2)} \, dt \ge \frac{c}{\sqrt{D}}.$$

Put $P(t) = |\sin(u_0 t)| F_{\mathcal{S}}(t)$ and $M = \max_{[t_-,t_+]} P$. Since $P \le F$ and $P \le M$, one has $P \le \sqrt{M}\sqrt{F}$. Equation (73), with the bounded kernel on this compact interval, therefore gives $M \ge \mu > 0$, independently of m, R. Choose t_* attaining this bound.

One retained orientation. For each pair,

$$A_i(t_*)B_i(t_*) \le 2^{-(r_i^0+r_i^1)} \le 4^{-R}, \qquad \min\{A_i(t_*), B_i(t_*)\} \le 2^{-R}.$$

Select the larger factor in every pair and call the resulting orientation s_*. Since all pair sums are at most one, telescoping products gives

$$P_{s_*}(t_*) := \prod_i \max\{A_i(t_*), B_i(t_*)\} \ge F(t_*) - m2^{-R} \ge \mu - m2^{-R}.$$

For $R \ge K_0 \log^2(m+2)$ and sufficiently large m, this is at least $\mu/2$. The orientation is retained: otherwise $F_{\mathcal{S}}(t_*) \le F(t_*) - P_{s_*}(t_*) \le m2^{-R} < \mu$, contrary to $P(t_*) \ge \mu$.

Let $a = \pi/(2t_*)$, in a fixed compact positive interval. Choose nearest integers k_j of odd parity on selected rails and even parity on unselected rails, and write

$$\frac{2u_j t_*}{\pi} = k_j + \delta_j.$$

Taking minus logarithms of $P_{s_*}(t_*) \ge \mu/2$, and using $-\log\cos x \ge x^2/2$, shows that

$$\sum_{\text{rail } j} \delta_j^2 \le K \tag{74}$$

for a fixed constant K. Thus the selected coordinates are close in total squared distance to an odd lattice, and the unselected ones to its even lattice.

Whole-set parity determinants. The norm bound and (74) also give

$$\sum_{\text{rail } i} k_i^2 \le K_1 D$$

for a fixed constant K_1. We use this total energy bound instead of bounding individual coordinates separately in each rail. Fix another orientation which switches exactly j pairs. At time $t > 0$, put $\gamma = t/t_*$ and choose nearest integers of the new required parities, writing

$$\gamma(k_i + \delta_i) = l_i + \epsilon_i, \qquad E = \sum_{\text{rail } i} \epsilon_i^2.$$

For disjoint sets of coordinates A, B such that every determinant $k_a l_b - k_b l_a$ is odd, substitution and squaring give

$$\begin{aligned} |A||B| &\le \sum_{a\in A, b\in B} (k_a l_b - k_b l_a)^2 \\ &\le 4\left[K_A E_B + K_B E_A + \gamma^2(K_A \Delta_B + K_B \Delta_A)\right] \\ &\le 4K_1 D(E + K\gamma^2), \end{aligned}$$

where $K_A = \sum_A k_i^2$, $E_A = \sum_A \epsilon_i^2$, and $\Delta_A = \sum_A \delta_i^2$. Indeed the determinant equals $\gamma(k_a\delta_b - k_b\delta_a) - k_a\epsilon_b + k_b\epsilon_a$.

If $j \le m/2$, take A to be the old selected coordinates in switched pairs and B the old selected coordinates in unchanged pairs. Their old parities are both odd, while their new parities are even and odd, respectively. Thus every determinant is odd, and $|A| \ge jR$, $|B| \ge (m-j)R$. If $j > m/2$, take A as before and B the old unselected coordinates in switched pairs. Their parity types are

odd-to-even and even-to-odd; again every determinant is odd, and both sets have size at least jR. In both cases

$$E \geq cjR - C\gamma^2.$$

Since t_* is bounded away from zero, there is a fixed $c_1 > 0$ such that $E \geq c_2 jR$ whenever $0 < t \leq c_1\sqrt{R}$. Each sine or cosine factor is $|\cos(\pi\epsilon_i/2)|$, with $|\epsilon_i| \leq 1$, and this is at most $e^{-\pi^2\epsilon_i^2/8}$. Hence the product of an orientation with j switches is at most $e^{-c_3 jR}$ throughout this time range. Summing by the number of switches gives

$$F(t) - P_{s_*}(t) \leq \sum_{j=1}^{m} \binom{m}{j} e^{-c_3 jR} = (1 + e^{-c_3 R})^m - 1 \leq Cme^{-c_3 R},$$

where the final inequality holds for $R \geq K_0 \log^2(m+2)$ and large m.

Between a fixed lower cutoff and $c_1\sqrt{R}$, the Fourier kernel has bounded total integral, so this contributes Cme^{-cR}. Below the cutoff, the earlier arithmetic–geometric-mean bound is exponentially small in D and is absorbed by this term. Above $c_1\sqrt{R}$, use the full-code tail from (73). This proves (68).

Multiplying by $w_{\max} \leq CD$ and dividing by $\|u\|_2 \geq c\sqrt{D}$ bounds the relative off-mass by

$$C\sqrt{D}\, me^{-cR} + C(1 + \sqrt{R})e^{-c\sqrt{R}}.$$

Here $D = mR$. If $R \geq K_L \log^2(m+2)$, then $\log m \leq \sqrt{R/K_L}$, while $\log R = o(\sqrt{R})$ uniformly as $m \to \infty$. Choose the fixed K_L large enough compared with L^2/c^2. The last exponential then absorbs every factor in $D^L(1 + \sqrt{R})$; the first term is smaller still. This proves (69), with the stated uniformity. □